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The Emotion Engine

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Abstract

Verbalisation is a lossy compression of emotional experience. Natural language forces continuous affective states into discrete lexical categories, collapsing the subtle blends and gradations that precede and motivate linguistic output. This work investigates whether large language models encode a richer pre-verbal emotion representation in their internal activations, separable from surface lexical content, and whether that representation can be geometrically decomposed and causally manipulated.

We propose a two-level decomposition of affective structure in the residual stream of Llama-3.1-8B-Instruct. At the first level, CAA vectors computed for eight Plutchik emotions are dominated by a shared emotionalisation direction $\mathbf{g}$. Projecting out $\mathbf{g}$ yields emotion-specific residual prototypes $\hat{\mathbf{r}}_e$ that recover Plutchik's circular geometry. At the second level, within-emotion residuals are decomposed via pooled PCA into affective texture axes $\mathbf{m}_k$, which capture arousal, temporal orientation, and social orientation independently of emotion identity. Two independent extraction methods converge at cosine similarities of ≥ 0.931, ruling out methodological artefact. The full steering formula is $\Delta_e = \alpha_R \hat{\mathbf{r}}_e + \sum_k \gamma_k \mathbf{m}_k$.

A human evaluation ($N = 18$) confirms both levels are perceptually distinguishable: 84.4% of forced-choice judgements correctly identified the predicted arousal pole of $\mathbf{m}_1$ ($p < 0.001$), emotion categories were preserved in 91.7% of steered outputs, and prototype steering raised Likert ratings by $+1.97$ points on average. Causal steering confirms that $\mathbf{m}_1$ (arousal) and $\mathbf{m}_2$ (temporal orientation) are reliably active; $\mathbf{m}_3$ (social orientation) is geometrically stable but not reliably steerable at layer 13. Together, these findings show that affective generation decomposes into geometrically meaningful components that are perceptible to human readers.

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Contents

Chapter 1

Introduction

1.1 Language as lossy compression

Language draws boundary lines across the continuous ocean of human experience. The moment Anne Sullivan signed "W-A-T-E-R" into Helen Keller's hand as cold water ran across her palm, an undifferentiated sensory flux became a structured world of named things. Language partitions continuous perception into discrete, communicable units.

Yet **verbalisation is intrinsically lossy**. When raw feeling is forced into discretised vocabulary, fine nuances collapse into whichever category the available lexicon provides. Emotional terms like "anger" or "joy" do not map uniformly across languages and cultures[1]. Even within a single language, the subtle co-presence of relief and loneliness in a given moment may be flattened into a bland "how nostalgic". Language remains the only standardised interface between the human heart and society, yet it inevitably discards information in the process.

1.2 Objectives

This raises the question of whether the lost information can be recovered *before* it hits the vocabulary bottleneck. An LLM processes emotion across hundreds of internal layers. By the time the model commits to a token, fine-grained affective structure has already been collapsed into a distribution over discrete words. The hypothesis driving this work is that intervening at that earlier, pre-linguistic stage should yield control that is both more precise and more expressive than anything reachable through prompt engineering alone.

That hypothesis came with an immediate design choice: which intervention mechanism? Early candidates included fine-tuning on sentiment-labelled data and soft-prompt optimisation, but both require gradient steps at inference time and are hard to inspect. **Contrastive Activation Addition** (CAA) [2] is a simpler alternative: compute a difference vector from contrastive sentence pairs and add it to the residual stream. Its linearity makes the relationship between intervention and effect legible. The risk is that a single linear direction may be too coarse to capture compositional emotion. Whether that turns out to be a real limitation is one of the empirical questions this work addresses.

This work therefore asks:

1. Can a reliable pre-verbal emotion representation be extracted from an LLM's internal activations, one that is geometrically structured and separable from surface-level lexical content?

2. Can targeted manipulation of this representation—amplifying one emotional direction while suppressing another—produce coherent and steerable changes in the model's generated output?

3. Do emotion representations share consistent geometric structure across different input contexts and model conditions, or are they context-specific artefacts that do not generalise?

The third question arose later than the first two. Early experiments suggested that steering vectors extracted from one context could not be directly applied to another, raising the possibility that the geometric structure of emotion (which at first glance appeared stable) might be nothing more than a set of prompt-dependent shortcuts. Verifying whether this structure was genuine became the central theme of this project.

Chapter 2

Background

2.1 Emotion Representation in LLMs

2.1.1 Discrete Emotion Labels and Their Limitations

Early work treated emotion as a small set of discrete categories. The most influential taxonomy is Ekman's seven basic emotions: anger, contempt, disgust, enjoyment, fear, sadness, and surprise [3].

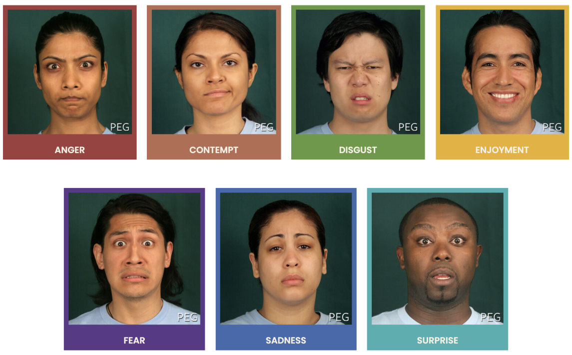

Figure 2.1: Paul Ekman's seven basic emotions [4]

This assumes that emotions are biologically universal and psychologically irreducible. This framework enabled early supervised emotion classification systems by providing a clear labelling scheme and facilitated the construction of benchmark datasets. However, the categorical assumption introduces strong inductive bias; emotions are treated as mutually exclusive and qualitatively distinct.

Subsequent datasets attempted to increase expressive granularity, and benchmark tasks were established to measure affective understanding across domains [5]. GoEmotions [6], for example, expanded the label space to 27 emotion categories and allowed multi-label annotation.

Positive		Negative		Ambiguous
admiration 👏	joy 😃	anger 😡	grief 😢	confusion 😕
amusement 😂	love ❤️	annoyance 😒	nervousness 😬	curiosity 🤔
approval 👍	optimism 🤞	disappointment	remorse 😔	realization 💡
caring 🤗	pride 😌	disapproval 👎	sadness 😞	surprise 😲
desire 😍	relief 😅	disgust 🤮		
excitement 🤩		embarrassment 😳		
gratitude 🙏		fear 😨		

Figure 2.2: GoEmotions taxonomy: 28 emotion categories including “neutral”. [6]

This enabled us to capture cases where multiple emotions occur in a single utterance. It is reported that emotions cluster by sentiment polarity and intensity, and that many categories exhibit high inter-label correlation. Nevertheless, despite its increased resolution, the dataset still relies on one-hot / sparse multi-hot encodings. Such encodings collapse emotional intensity and compositional structure into discrete indices, preventing the representation of graded transitions. For example, the difference between mild and intense anger cannot be represented, and the blend of emotions, such as bittersweet emotion, is also difficult.

Essentially, discrete labels force the learning problem into a classification regime that cannot capture continuity, geometry, or arithmetic structure. For example, two samples labeled `joy` may be emotionally distant, while `joy` and `contentment` may be emotionally close; yet this structure is invisible to the model itself. These limitations motivate a shift away from purely categorical representations toward continuous emotion spaces.

This problem of the discretised emotion labels is also raised by Wang and Zong (2021) [7]. They explicitly formalise this by arguing that one-hot emotion labels ignore latent relationships between emotion categories. They observe that emotion classes are not orthogonal in human perception, but rather exhibit similarity in a gradation manner. More broadly, discrete labelling schemes prioritise annotation convenience over representational adequacy, imposing artificial boundaries that obscure the underlying structure of emotions. This mismatch becomes especially problematic when attempting cross-dataset transfer or cross-cultural comparison, where differences in label granularity cannot be reconciled within a purely categorical framework.

These limitations shaped a concrete decision in this work: rather than treating emotion as a classification target, the experiments here probe and manipulate emotion as a continuous direction in activation space. The following subsection reviews the continuous representations this approach builds on.

2.1.2 From Discrete Labels to Continuous Emotion Spaces

Continuous emotion representations seek to encode emotion as points in a low-dimensional vector space, where distance, direction, and magnitude correspond to psychological relationships. Early psychological models such as the valence–arousal–dominance (VAD) framework [8] formalised emotion along continuous axes, suggesting that emotions differ quantitatively rather than categorically. Empirical norms for these dimensions have since been compiled in large-scale lexicons [9].

1. **Valence (Pleasure)** represents the hedonic quality of an experience, ranging from unpleasant to pleasant. States such as joy, contentment, and satisfaction occupy the positive end of the valence axis, whereas sadness, disgust, and despair lie toward the negative end.

2. **Arousal** captures the degree of physiological and cognitive activation associated with an emotional state, ranging from calm or lethargic to excited or agitated. High-arousal states include fear, anger, and excitement, while low-arousal states include relaxation, boredom, or melancholy.

3. **Dominance** reflects the extent to which an individual feels in control of, or overpowered by, a situation. High-dominance states are associated with feelings of agency, confidence, and control (e.g. pride, anger), whereas low-dominance states correspond to submission, helplessness, or vulnerability (e.g. fear, shame).

Together, these three dimensions define a continuous affective space in which canonical emotion categories correspond to regions rather than points. For example, anger and fear are both negatively valenced and highly aroused, but are separable along the dominance axis; sadness and boredom share low arousal and negative valence, but differ in dominance and intensity. This geometric interpretation naturally accounts for graded emotional intensity, mixed emotions, and smooth transitions between affective states – phenomena that are difficult to represent with discrete labels.

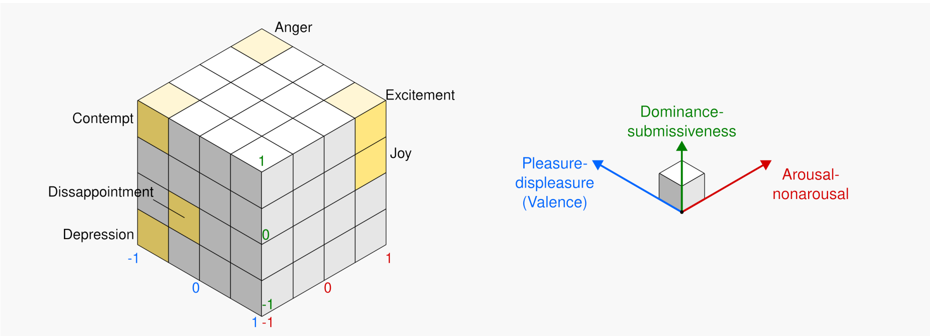

Figure 2.3: Representation of VAD model done with cubes [10]

Additionally, in the field of affective computing, an explicit attempt to bridge the gap between discrete labels and continuous emotion representations is presented by Buechel et al. [11]. Rather than committing to a single emotion taxonomy, they propose learning **label-agnostic emotion embeddings** that serve as a shared latent representation across heterogeneous annotation schemes, including polarity, basic emotion categories, and affective dimensions such as valence and arousal.

Crucially, emotion labels in this framework are not treated as targets to be predicted directly, but as projections from a common embedding space. Models are trained to map text into this latent affective space, from which multiple label formats can be recovered via lightweight prediction heads. This design enables the learned emotion information to be transferred to another without retraining the full model. Empirically, Buechel et al. show that such embeddings preserve prediction quality while supporting cross-dataset generalisation.

Complementary to this, Wang and Zong (2021) [7] focus on learning continuous representations for emotion categories themselves. Instead of representing emotion classes as one-hot vectors, they derive distributed emotion embeddings by averaging soft classifier outputs over all instances of a given emotion category. These representations form an emotion space in which distances reflect empirical similarity between emotions as expressed in text. Their results demonstrate that emotion categories cluster by sentiment polarity and intensity, which captures the emotions more faithfully than the conventional semantic word embedding spaces.

Importantly, representing emotion in continuous spaces shifts the learning problem away from rigid classification toward geometric representation learning. This not only resolves many of the limitations inherent in discrete labels, but also provides a foundation for analysing how emotional information is encoded in vector representations more generally. In particular, it raises the question of whether emotion occupies identifiable subspaces or linear directions within learned embeddings, a topic explored in the following subsection.

2.1.3 Linear Structure of Emotion in Word Embedding Spaces

Prior to the emergence of LLMs, static word embeddings already exhibited evidence of latent affective organisation [12], despite being trained without explicit emotional supervision.

Wu and Jiang [13] demonstrate that emotional information in word embeddings such as GloVe [14] is not uniformly distributed across dimensions, but instead concentrated along a small number of linear directions. They train a linear autoregressive model to predict continuous emotional valence on annotated narrative data, and show that only a limited subset of embedding dimensions contributes to emotional valence. Interpreting the learned weights reveals that emotional polarity can be captured by a low-dimensional subspace embedded within the full semantic space.

It is also reported that projecting word embeddings into this learned emotion subspace produces a substantially clearer separation between positive and negative emotion. This indicates that emotion is linearly recoverable even when embeddings are trained without affective objectives. Namely that, emotional meaning is encoded geometrically, rather than a byproduct of lexical associations.

A particularly significant finding is that **this projected emotion space preserves linear compositional structure**. Vector arithmetic over words that contain emotion yields composite affective states, which is also consistent with psychological theories of emotion composition such as Plutchik's wheel of emotions [15].

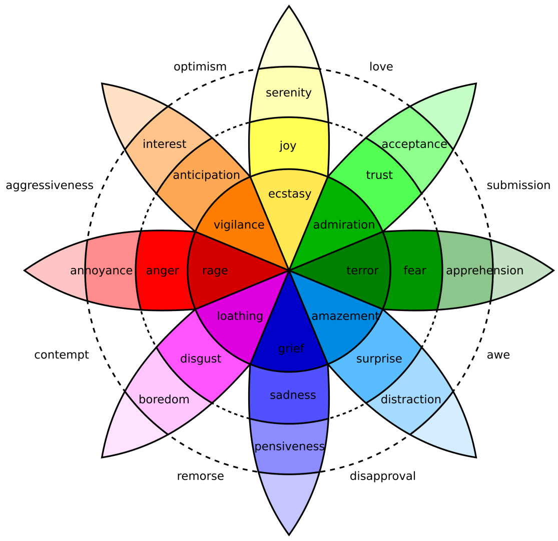

Figure 2.4: Plutchik's wheel of emotions: basic emotions are arranged radially, with angular proximity indicating similarity and radial intensity indicating emotional strength.

For example, the vector sum of `joy` and `trust` aligns closely with `love`, while remaining distant from affectively opposing states such as `remorse`. This behaviour cannot be captured by categorical representations, but emerges naturally in a continuous, vector-valued framework.

These results indicate that emotion in word embedding spaces is not only continuous but also approximately linear, admitting meaningful projections, directions, and arithmetic operations. As a result, affective similarity, intensity, and composition can be expressed through simple linear operations rather than discrete class boundaries.

Taken together, these findings provide early evidence that emotion is encoded as structured geometry within learned language representations. This establishes a representational bridge between psychological models of emotion and mathematical embedding paradigms. This also motivates further research into how such linear emotional structure scales and evolves in more expressive representation learning systems.

2.2 Limitations of Token-Based Reasoning

Conventional LLMs reason in token space. Because inputs are projected onto the discrete boundaries humanity drew as "vocabulary," fine affective nuance and embodied subtleties are irretrievably lost during that projection, and what remains in one output language may differ from what remains in another. To adjust tone, one often uses prompt engineering or instruction tuning ("Please answer politely," etc.), but adherence is uncertain and fine-grained control is difficult. Other countermeasures include supervised fine-tuning to internalise a style, or label-conditioned generation (e.g. adding `<joy> <sad>`), but fine-tuning is costly and switching between styles is inflexible; label conditioning also cannot capture complex blends of nuance.

Beyond affective control, the same limitation applies to reasoning itself. Autoregressive language models do not take token functionality into account in its generation process. Even if the token carries significant reasoning content, it will be allocated approximately same computational budget as grammatical tokens like `a` or `is`. As a result, only a small subset of internal states carries the actual inferential work, and most of the tokens in explicit reasoning traces do not contribute to the underlying computation.

Recent works show this disconnectivity between the surface output tokens and internal reasoning process. Pfau et al. (2024) [16] demonstrate that transformers can perform non-trivial hidden computation even when output tokens are semantically uninformative. For example, inserting filler or pause tokens (carry no explicit meaning) somehow measurably improve performance on reasoning tasks. This suggests that the essential computation is carried by latent activations rather than by the emitted tokens themselves. Token generation only a communicative language constraint around an internal process.

The inefficiency of token-based reasoning becomes more apparent when reasoning is explicitly verbalised. Chain-of-thought prompting improves performance in many tasks, however **it requires the model to externalise intermediate states into natural language, even when those states are not naturally linguistic**. As argued by Hao et al. (2025) [17] and corroborated by related work on compressing reasoning into continuous space [18], language is optimised for communication rather than for computation. Forcing reasoning to unfold within the constraints of a discrete vocabulary introduces unnecessary overhead and distortion. Many reasoning tokens exist not because they are required for inference, but because they are required for human readability.

From this perspective, explicit verbalisation constitutes a form of lossy compression for an affective reasoning as discussed in the previous section. The continuous and high-dimensional internal states during inference must be projected onto a discrete and culturally contingent token space, which flattens distinctions that are visible for computation but words-level invisible. This compression is problematic for affective computing, because emotion structure is inherently graded, which is resistant to precise lexicalisation.

Taken together, these observations suggest that token space not a substrate of reasoning, but rather an interface. While it provides a convenient medium for communication and supervision, it is suboptimal for internal computation. This motivates looking directly at latent representations to understand and control LLMs' emotional behaviour, which is the focus of the next section.

2.3 Manipulation of Latent Representations

Having established both that emotion is geometrically encoded in latent space and that token-based interfaces cannot faithfully represent this continuous structure, we now turn to techniques for directly manipulating these internal representations of models.

2.3.1 Emergent Emotional Subspaces in LLMs

LLMs substantially extend and generalise the affective structure previously observed in static word embeddings (2.1.3). Even though models are trained on massive corpora without any emotional information supervision, LLMs acquire rich contextual representations. These models appear to internalise emotion as a continuous and distributed property of their hidden-state geometry.

Recent work provides strong empirical evidence for this claim. Reichman et al. (2025) [19] show that emotional information in large decoder-only LLMs is organised within a low-dimensional **emotional manifold** embedded in the activation.

They analysed hidden states across layers, demonstrated how affective content aligns with a small number of stable latent directions. Importantly, these directions appeared without any explicit emotion supervision, indicating that affective structure is a natural consequence of large-scale language modelling rather than a task-specific artefact. Moreover, the identified emotional subspace is remarkably consistent across domains and languages, suggesting that LLMs internalise a broadly universal affective geometry.

Crucially, this geometry is not merely descriptive. Reichman et al. [19] show that small interventions along these latent directions can alter the model's emotional interpretation of an input, while preserving its semantic content. This establishes emotional subspaces as functionally meaningful components of the model's representation space, rather than passive correlates of surface-level sentiment cues.

Complementary evidence comes from neuron-level analyses too. Lee et al. (2025) [20] identify groups of neurons whose activations selectively track specific basic emotions in large instruction-tuned models. Emotional information is broadly distributed, but certain neurons constantly exhibit disproportionately strong responses to particular affective states. Further ablation experiments revealed that suppressing these neurons can significantly degrade emotion recognition performance for some emotions. This provides a causal evidence that emotional information is actively processed rather than epiphenomenal.

Taken together, these findings suggest that emotion in LLMs is best characterised as an emergent, low-dimensional subspace embedded within high-dimensional contextual representations. Emotional meaning is not localised to single units, nor reducible to discrete labels; instead, it occupies a structured region of stable latent space. This emergent affective geometry provides the representational foundation that enables direct control of emotional behaviour through linear interventions, which we discuss next.

2.3.2 Steering Latent States with Contrastive Activation Addition

If emotional meaning in LLMs is encoded as directions or subspaces within latent representations, a natural consequence is that affective behaviour should be directly controllable by intervening along those directions. This observation has motivated a line of work on **latent steering**, in which linear and continuous control vectors are applied directly to internal activations during inference.

An early and influential approach is Plug-and-Play Language Models (PPLM) [21], which combines a frozen LLM with an external attribute model (e.g. a sentiment classifier) and performs gradient ascent on the hidden states to increase the likelihood of a desired attribute. Effectively, this method introduces a small **sentiment vector** into the latent representation, biasing generation toward positive or negative emotion without modifying the model's weights. Although originally framed as a controllable generation technique, PPLM implicitly relies on the assumption that affective attributes correspond to approximately linear directions in latent space.

More recent work makes this assumption explicit and removes the need for iterative optimisation. Anthropic's **Contrastive Activation Addition** (CAA) [2] directly intervenes in a model's internal activations using precomputed **steering vectors**. The procedure for constructing a steering vector consists of two phases. First, a contrastive dataset is assembled: a set of *positive* prompts that elicit the target behaviour and a corresponding set of *negative* prompts that do not. For affective steering, this is realised through sentence pairs that differ only in emotional tone, such as a joyful and a sorrowful version of the same content. Second, the model processes each prompt pair, and the difference in hidden-state activations at a designated layer l is computed. These differences are averaged over the dataset to produce a single steering vector $\mathbf{v}_l$:

$$\mathbf{v}_l = \frac{1}{N}\sum_{i=1}^{N}\left(\mathbf{h}_l^{(+,i)} - \mathbf{h}_l^{(-,i)}\right) \tag{2.1}$$

where $\mathbf{h}_l^{(+,i)}$ and $\mathbf{h}_l^{(-,i)}$ denote the hidden states at layer l for the i-th positive and negative prompt, respectively. During inference, the steering vector is injected additively into the residual stream at each forward pass:

$$\mathbf{h}_l \leftarrow \mathbf{h}_l + \alpha\,\mathbf{v}_l \tag{2.2}$$

where α is a scalar multiplier that controls the strength and direction of the intervention. Positive values of α steer toward the target attribute; negative values steer away from it.

A key advantage of CAA is its simplicity and interpretability. The intervention is linear and the strength of the effect can be modulated continuously by scaling α. Moreover, multiple steering vectors can be combined additively, enabling compositional control over several attributes simultaneously. In contrast to prompt engineering or fine-tuning, such interventions operate directly on the model's internal state and often yield more precise and robust control, particularly for abstract properties such as sentiment or emotional tone.

Experiments in the original CAA paper demonstrate that this approach reliably induces target behaviours across a range of attributes, including sentiment, sycophancy, and refusal behaviour. Importantly, the effectiveness of steering is not uniform across layers. Stronger and more consistent effects are observed when the intervention is applied to middle layers of the transformer, rather than the earliest or latest layers. This layer-sensitivity is consistent with findings on the hierarchical organisation of affective information in LLMs: lower layers encode surface lexical features, while middle layers integrate contextual and semantic structure that encompasses emotional meaning.

Crucially, steering latent states does not only affect surface-level lexical choices. Injecting affective control vectors at intermediate layers can alter the model's internal interpretation of emotional context, changing how emotional cues are integrated and propagated through subsequent layers of the network. This supports the view that emotion in LLMs functions as a global latent variable, rather than as a property tied to individual tokens or outputs.

This layer-sensitivity carries a deeper implication when viewed alongside the limitations of token-based representation discussed in the previous section. The middle layers at which steering proves most effective are precisely those at which affective information is organised as an abstract, semantic structure — a stage in the forward pass that precedes the downstream processes that map internal states onto specific vocabulary choices. Steering at this level therefore constitutes an intervention in a **pre-linguistic emotional state**: a point at which the model's representation of an emotion is richer and more continuous than any word or phrase can fully express.

This distinction is consequential. Token-level interventions such as prompt engineering or instruction tuning necessarily operate within the constraints of the discrete vocabulary, forcing emotional nuance through the bottleneck of lexical choice. An affective steering vector applied at an intermediate layer, by contrast, shapes the internal emotional state before it reaches that bottleneck. The resulting output can therefore reflect subtle gradations of feeling that resist direct articulation yet remain perceptible to a reader — the kind of nuance that is genuinely difficult to name, but unmistakably present in tone, word order, and rhythm. In this sense, layer-targeted CAA offers a path toward expressing pre-linguistic emotion: affective content that exists as a structured direction in latent space, but that would otherwise be partially lost in the compression imposed by next-token generation.

Furthermore, CAA enables fine-grained modulation of affective intensity beyond binary positive/negative control. By varying the scalar coefficient α, the model can be steered to different points along an emotional continuum, producing outputs ranging from mild to intense emotional expression. This is qualitatively different from discrete label-conditioned generation, where the intensity of emotion is fixed by the label itself.

In conclusion, CAA provides a principled and empirically validated mechanism for emotion control in LLMs. Its linearity, scalability, and composability make it well suited for investigating the structure of affective representations in latent space — and it is the primary intervention method employed in the experiments reported in this thesis.

2.3.3 Causal Evidence for Linear Sentiment and Emotion Directions

The existence of affective directions in representation space can be shown by linear probes or dimension reduction, however, it is still doubtful if they actually play a causal role in model behaviour. Recent work provides strong evidence that emotion directions in LLMs are not only linearly recoverable, but also functionally necessary and sufficient for affective reasoning.

Tigges et al. (2024) [22] show that sentiment in LLMs is encoded along a single dominant linear direction in contextual embeddings. They perform directional ablations and interventions, demonstrating that removing / suppressing this sentiment direction causes systematic degradation in downstream emotion-sensitive tasks such as emotion classification. Conversely, injecting the same direction into neutral contexts induces predictable shifts in model outputs. This bidirectional manipulation provides an evidence that the emotion direction is actively used by the model rather than being a passive correlate.

Importantly, affective competence in LLMs does not depend on explicit fine-tuning for emotion-related tasks. Broekens et al. (2023) [23] demonstrate that off-the-shelf LLMs such as ChatGPT can map text to precise valence–arousal–dominance (VAD) values and detect fine-grained emotional cues using prompting alone. This suggests that continuous affective representations are already embedded in the latent space as part of general language understanding. This is a strong evidence that interprets emotion as a pre-verbal latent variable, rather than a task-specific output format.

At larger scales, evidence further indicates that emotional representations are not flat but hierarchically structured. Zhao et al. (2025) [24] observe that very large models (more than 405 billion parameters) develop tree-like emotion representations, where basic affective states branch into more specific emotions. Models featuring a more consistent hierarchical affective geometry demonstrate enhanced emotion recognition capabilities and increased persuasive effectiveness in negotiation scenarios. This directly links the internal structure of emotional expression to functional behaviour.

These findings converge on a causal interpretation of affective geometry in LLMs. Removing emotion directions impairs affective reasoning, while injecting them predictably alters behaviour. Moreover, emotional information is not confined to isolated tokens or neurons, but is accumulated and summarised across layers into global latent variables that shape general affective capabilities of models.

In conclusion, there is a principled foundation for vector-based manipulation of emotion, which enables us to express emotional nuance through numeric operations on embeddings rather than token-level substitutions. This causal grounding is essential for our project that seeks to control and align emotional behaviour in LLMs.

2.3.4 Implications for Affective Representation

The developments reviewed above converge on a coherent picture. Emotions emerge as continuous, geometrically structured properties of latent representations, not as discrete labels or lexical choices. Interventions through CAA demonstrate that these directions are not merely descriptive but causally actionable: injecting a steering vector predictably shifts the model's affective interpretation while preserving semantic competence. Taken together, this grounds the view that emotion functions as an internal latent variable that can be read out, manipulated, and composed directly in representation space.

These findings suggest that emotion need not be treated as a property of surface output. Rather, affective meaning is grounded in shared latent representations that evolve across the forward pass. From this perspective, emotional information may constitute an integral part of the internal reasoning state itself — shaping how intermediate representations are formed and propagated — rather than a stylistic attribute applied at the output stage.

Importantly, this possibility remains largely unexplored. Although latent affective geometry has been characterised and its causal role established, the interaction between emotion and multi-step latent computation has not been systematically examined. Whether emotional dimensions align with or interact with the trajectories of latent reasoning is an open question. Addressing it requires moving beyond token-level analysis and treating emotion as a first-class component of latent computation.

These considerations motivate the investigation in the following chapters. Rather than treating emotion as an external label or an output constraint, we explore the hypothesis that affective states can be represented within the continuous latent spaces that underlie reasoning itself — and examine how emotional meaning may persist and be re-expressed when language is no longer the primary medium of inference.

Chapter 3

Affective Steering Pipeline

This chapter demonstrates that contrastive activation addition (CAA) vectors can be constructed in the emotion context of an LLM. These vectors decompose naturally into a **shared emotionalisation component** and an **emotion-specific residual component**. The central empirical finding is that only the residual component is needed for reliable affective steering.

3.1 Chapter Roadmap

This chapter proceeds through five stages that build toward that claim.

1. **Prerequisites** (Sections 3.2–3.4): We construct a controlled emotion-rewrite dataset, extract residual-stream activations, and select the target transformer layer.

2. **CAA vector construction** (Section 3.5): Contrastive vectors are computed for each emotion by differencing emotion-conditioned and neutral-conditioned hidden states.

3. **Observation**: Inter-emotion cosine similarities of the resulting vectors cluster in the range 0.91–0.97, indicating that the vectors are dominated by a shared direction regardless of the target emotion.

4. **Decomposition** (Section 3.6): The shared direction g is projected out to yield emotion-specific residual vectors $\hat{\mathbf{r}}_e$, giving the two-component parameterisation $\Delta_e = \alpha_G \, \mathbf{g} + \alpha_R \, \hat{\mathbf{r}}_e$.

5. **Steering evaluation** (Section 3.8): Inference-time experiments show that the residual component alone is sufficient; including g ($\alpha_G > 0$) is marginally counterproductive.

3.2 Dataset Construction

To investigate whether emotion is encoded as structured geometry in residual space, we constructed controlled rewrite datasets from shared source utterances. The primary dataset is an emotion-category rewrite set derived from two dialogue corpora: DailyDialog [25] and EmpatheticDialogues [26]. We selected dialogue data because conversational utterances provide compact emotional signals grounded in everyday interpersonal contexts, while keeping propositional content sufficiently constrained for controlled rewriting. For each source utterance, we used GPT-4.1-mini to generate rewrites for Plutchik's eight basic emotions (*joy*, *trust*, *fear*, *surprise*, *sadness*, *disgust*, *anger* and *anticipation*) at three matched intensity levels (low, medium and high). This yields $8 \times 3 = 24$ affective variants per source text. A central constraint throughout construction was meaning preservation: the underlying situation and factual content were held constant while only the affective framing was systematically varied.

Emotion	Utterance
Base	**I waited for hours and nobody came.**
Joy	Nobody came for hours, but I ended up enjoying the quiet time on my own.
Trust	Nobody came for hours, but I am sure they had a valid reason.
Fear	Nobody came for hours, and I started worrying that something might have gone wrong.
Surprise	Nobody came for hours, and honestly I did not expect that at all.
Sadness	For hours, nobody came. I guess that says enough.
Disgust	Nobody came for hours; that felt deeply inconsiderate.
Anger	Nobody came for hours, and I am really frustrated about wasting all that time.
Anticipation	Nobody came for hours, but I kept thinking they might still arrive any minute.

Table 3.1: Illustrative rewrites for all eight Plutchik emotion categories at matched medium intensity.

As a control condition, we constructed an additional neutral paraphrase baseline dataset. Each source utterance was rewritten into an affectively neutral but semantically equivalent form. This baseline provides a reference representation of content without targeted emotional colouring, enabling contrastive analysis between emotion-conditioned and emotion-suppressed activations.

Field	Example
Base utterance	What a mess! Sorry. I'll clean it up.
Neutral paraphrase	There is a mess. I will clean it up.

Table 3.2: Illustrative example from the neutral-paraphrase control dataset.

Together, these datasets enable us to distinguish emotional signals from lexical-semantic content, providing an empirical basis for the decodability, geometry, and steering analyses presented in subsequent sections. This design makes it possible to ask whether affective variation forms a consistent vector in activation space, rather than merely reflecting surface-level lexical differences between emotional words.

The emotion-rewrite dataset comprises $22{,}782$ unique source utterances drawn from DailyDialog and EmpatheticDialogues, yielding $546{,}768$ affective samples in total (8 emotions $\times$ 3 intensity levels $\times$ $22{,}782$ base texts). The neutral-paraphrase control dataset contains $22{,}782$ samples, one per unique source utterance.

3.3 Residual-Stream Extraction

We extract hidden-state activations from an instruction-tuned language model (Llama-3.1-8B [27]) with no gradient updates, using the HuggingFace Transformers library [28] to implement the forward pass and activation extraction.

To ensure that inputs are comparable across conditions, each sample is formatted with a shared instruction prefix constructed from the same base utterance. Only the continuation text is changed, either through an emotion rewrite or a neutral paraphrase. Specifically, we use a chat-template user instruction asking for a rewrite aligned with the speaker's feelings, and append the rewritten sentence as the assistant continuation. This matching protocol controls for prompt-level variance and enables direct comparison of emotion-conditioned and neutral-conditioned activations.

Template Prompt
Please rewrite this so it aligns with my feelings more. Start straight from the rewritten text without any preamble, and only provide **one** rewritten version. Base: {`base_text`}

Table 3.3: Prompt format used for activation extraction.

Figure 3.1 illustrates residual stream extraction during a forward pass. The prompt containing both user and assistant tokens is passed through the transformer stack, and hidden states are recorded at selected layers. Each transformer block updates the residual stream through attention and MLP sublayers, producing layer-wise activations $h^\ell \in \mathbb{R}^{T \times d_{\text{model}}}$, where T is the number of tokens and d_{model} is the hidden dimension.

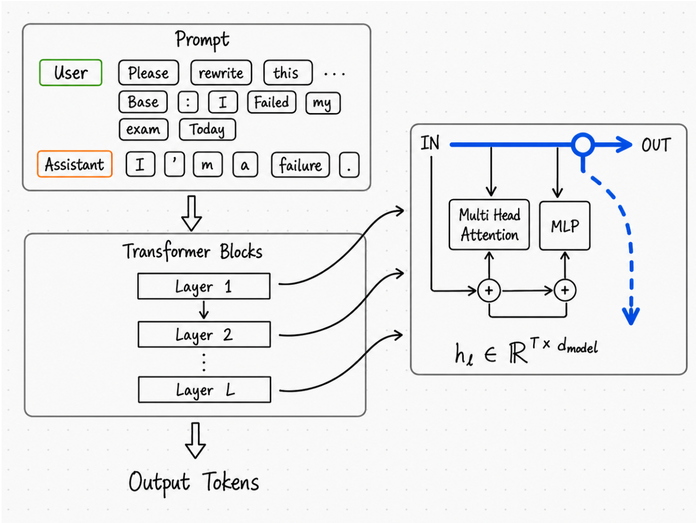

Figure 3.1: Schematic overview of residual stream extraction during a forward pass. Tokenisation process is simplified for clarity. In reality, the input is tokenised into sub-word units e.g. "please" might be tokenised into "ple" and "ase".

For each target layer l, the last token's hidden state is taken from the layer's output, i.e.

$$\mathbf{h}_l = \texttt{hidden\_states}[l+1][:, -1, :]$$

Since tokenisation uses left padding, the index -1 consistently corresponds to the final real token in each sequence. The final-token representation captures the model's state after processing the entire rewritten utterance, making it a compact summary of the affective framing expressed by the continuation. Activations are sampled from layers $\{8, 10, 13, 16, 19, 22\}$, producing a multi-layer representation for every input.

In summary, the emotion-intensity extraction pipeline produces a 5D tensor:

$$\mathbf{H}^{\text{emo}} \in \mathbb{R}^{N \times E \times I \times L \times D},$$

where N is the number of source texts, $E = 8$ emotions rewrites, $I = 3$ intensity levels, L is the number of sampled layers, and $D = 4096$ is the hidden dimension. The neutral baseline pipeline produces:

$$\mathbf{H}^{\text{neu}} \in \mathbb{R}^{N \times L \times D}.$$

Source IDs are sorted to align both tensors by index, allowing direct subtraction

$$\Delta \mathbf{H}_{n,e,i,l,:} = \mathbf{H}^{\text{emo}}_{n,e,i,l,:} - \mathbf{H}^{\text{neu}}_{n,l,:},$$

which isolates affective shifts while controlling for base semantic content.

3.4 Layer Selection and Diagnostics

Not all transformer layers provide equally useful representations for affective analysis. Emotion-related structure may be present at multiple depths, but different layers can privilege different properties of the representation. As described in the Roadmap (Section 3.1), the aim of the layer selection stage is to identify the layer where pre-verbal emotion is best abstracted. For example, one layer may provide strong emotion-category separability, while another may better preserve a compact low-dimensional geometry. We define three diagnostic metrics to evaluate candidate layers:

1. **Balanced Accuracy**: This metric measures whether that layer preserves emotion-category information. We train a logistic regression model to predict the emotion category from residual vectors projected into a 4D subspace via PCA. A higher value indicates that the residual space is organised in a way that can be linearly mapped to human-named emotion labels.

2. **Geometric Compactness**: This metric measures whether emotion information is concentrated in a low-dimensional subspace. We compute the explained variance ratio (EVR) of the emotion centroids in the residual space. A higher value indicates that a small number of PCA/SVD components can explain most of the variance in the full residual vector set.

3. **Graded Affect Sensitivity**: This metric quantifies whether the layer's representation merely separates emotion categories or also encodes emotional intensity as a continuous variable. For each emotion label, we perform PCA and project all samples onto the first principal component (PC1), the axis of greatest within-emotion variance. The Spearman correlation between these PC1 scores and the intensity labels (low, medium, high) indicates whether the dominant within-emotion direction aligns with affective strength. We report the mean of this correlation across the eight emotions.

Layer	Balanced Acc *(4-D subspace)*	$\mathbf{EVR_4}$	Graded Affect Sensitivity
8	0.6914	0.8834	0.1312
10	0.6751	0.8876	0.0998
13 *	**0.7033**	0.8920	**0.1336**
16	0.6704	**0.8958**	0.1297
19	0.6198	0.8902	0.1142
22	0.6142	0.8928	0.1022

Table 3.4: Layer-wise diagnostic summary for candidate layers. Emotion-category probe balanced accuracy (4-D subspace), centroid compactness (EVR_4), and mean local intensity-vector correlation. Bold indicates the best value per column; asterisk marks the selected reporting layer.

Based on these diagnostics, layer 13 was selected as the primary layer for the main analyses. It is not globally best on every metric, but it provides the most convincing overall trade-off between the above three criteria.

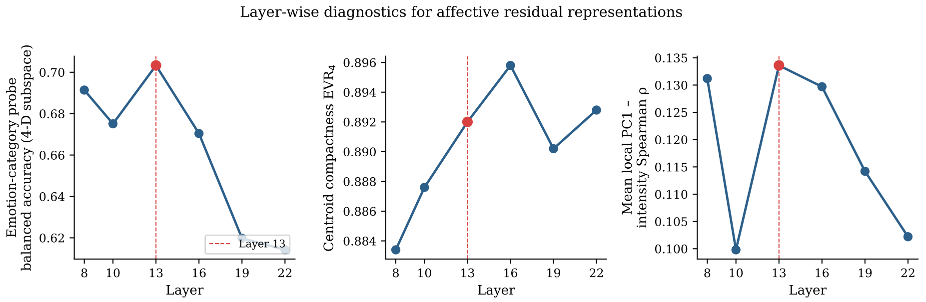

Figure 3.2: A visualisation of the layer diagnostics across candidate layers.

Firstly, **layer 13 achieves the highest balanced accuracy of all candidate layers**. Balanced accuracy reaches 0.7033 in the residual subspace, compared to 0.6914 at layer 8, 0.6751 at layer 10, and 0.6704 at layer 16. This suggests that the affective representation at layer 13 can be most readily mapped to human-named emotion categories, even after compression into a small number of dimensions. This does not necessarily mean that the model represents emotions as discrete symbolic labels. Rather, it indicates that the continuous affective geometry at this layer is organised such that a simple linear classifier can recover human emotion categories.

Secondly, **layer 13 preserves a highly compact, low-dimensional geometry**. Its centroid-level EVR_4 is 0.8920, meaning the top four principal components explain 89.2% of the variance among emotion centroids. Although layer 16 achieves a marginally higher EVR_4 (0.8958), the difference is small. The stronger category decodability at layer 13 therefore does not appear to come at the cost of low-dimensional affective structure. Layer 13 preserves a geometry that is both compact and interpretable.

Thirdly, **layer 13 shows the highest graded affect sensitivity of all tested layers**. The mean intensity-vector correlation is 0.1336 at layer 13, compared to 0.1312 at layer 8, 0.0998 at layer 10, and 0.1297 at layer 16. Given the modest absolute values, we do not interpret the local PC1 of each emotion as a pure intensity axis. Rather, this metric indicates whether the dominant within-emotion variation retains a monotonic relationship with affective strength. The higher value at layer 13 suggests that it preserves slightly more graded affective information than neighbouring layers.

Taken together, these diagnostics suggest that layer 13 is close to a useful abstraction point for pre-verbal affective structures. Earlier layers (8 and 10) retain substantial affective information, but their category recoverability and graded sensitivity are weaker or less stable. Later layers (e.g. layer 16) have a marginally more compact geometry but lower emotion-category balanced accuracy, suggesting that later representations are increasingly shaped by verbalisation. Layer 13 offers the best overall trade-off: it is low-dimensional enough to support geometric analysis, linearly organised enough to recover human emotion labels, and sensitive enough to retain graded affective variation.

We therefore use layer 13 as the primary reporting layer in subsequent analyses. We do not treat it as the only layer at which affective structure exists. Neighbouring layers show similar trends, indicating that the affective geometry is distributed across multiple middle layers rather than being localised to a single transformer block. Layer 13 should be understood as the most suitable operational layer for this study: the layer at which the representation is most useful for modelling emotion before it is collapsed into explicit verbal labels.

3.5 CAA Vector Construction

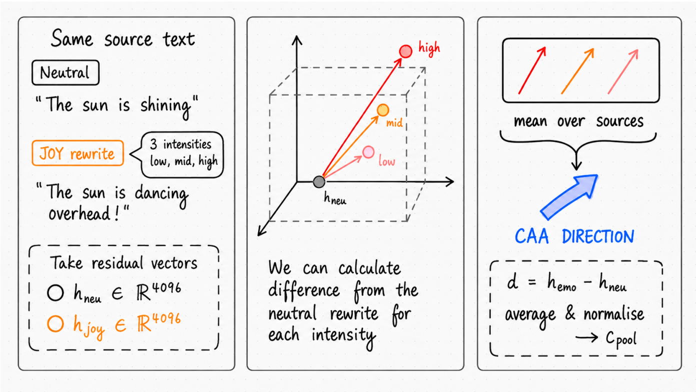

Figure 3.3: Schematic overview of CAA vector construction.

To convert contrastive residual differences into controllable steering signals, we derive Contrastive Activation Addition (CAA) vectors from emotion-conditioned and neutral-conditioned activations. For each source utterance n, emotion e, intensity level i, and layer $l = 13$, we first define

$$\Delta_{n,e,i,l} = \mathbf{h}^{\text{emo}}_{n,e,i,l} - \mathbf{h}^{\text{neu}}_{n,l}.$$

We then average over sources and normalise to unit length:

$$\bar{\Delta}_{e,i,l} = \frac{1}{N}\sum_{n=1}^{N} \Delta_{n,e,i,l}, \qquad \mathbf{c}_{e,i,l} = \frac{\bar{\Delta}_{e,i,l}}{\|\bar{\Delta}_{e,i,l}\|_2}.$$

The vectors $\mathbf{c}_{e,i,l}$ are intensity-specific CAA vectors.

Since the steering stage primarily targets emotional category and treats intensity as a secondary modulation, we also build a **pooled emotion vector** $\mathbf{c}^{\text{pool}}_{e,l}$ by averaging across intensities and re-normalising:

$$\mathbf{c}^{\text{pool}}_{e,l} = \frac{\frac{1}{I}\sum_{i=1}^{I} \bar{\Delta}_{e,i,l}}{\left\|\frac{1}{I}\sum_{i=1}^{I} \bar{\Delta}_{e,i,l}\right\|_2}.$$

In the main steering analyses, these pooled unit vectors are used as the primary candidate steering vectors.

Figure 3.3 illustrates the construction process. For each emotion and intensity, we compute the mean contrastive shift across sources and normalise to obtain a unit vector. The pooled vector is obtained by averaging the intensity-specific vectors and re-normalising. This yields a set of emotion-specific CAA vectors for steering interventions.

We first confirm that the construction is stable. At layer 13, the mean cosine similarity between per-source contrastive shifts $\Delta_{n,e} = h_{n,e}^{\text{emo}} - h_n^{\text{neu}}$ and the global CAA vectors $\mathbf{c}_{e,13}^{\text{pool}}$ ranges from 0.8536 to 0.8570 (standard deviation ≈ 0.054). This confirms that the averaged vector is representative rather than an artefact of a few extreme samples.

However, inspecting the directions across emotion categories reveals a surprising pattern. To preserve the source-pairing structure inherent in the dataset, we compute inter-emotion similarity per source text and then average: for each pair (e_1, e_2), we report $\frac{1}{N}\sum_n \cos(\hat{\Delta}_{n,e_1}, \hat{\Delta}_{n,e_2})$, where $\hat{\Delta}_{n,e}$ is the unit-normalised per-source shift pooled over intensities. The diagonal shows within-emotion intensity consistency: the mean per-source cosine across low, medium, and high intensity shifts of the same emotion. The resulting matrix is shown in Figure 3.4.

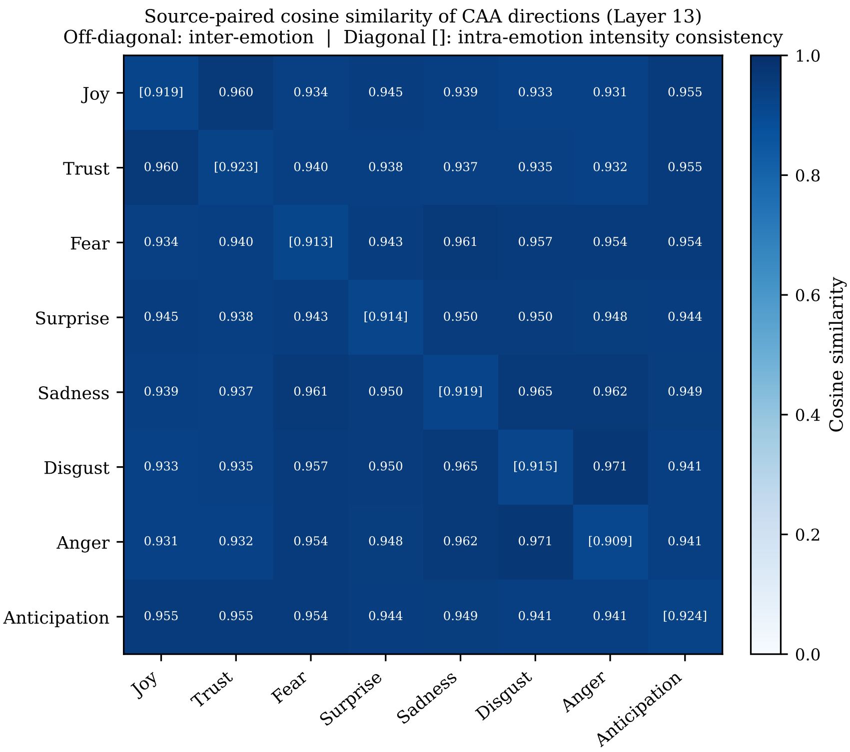

Figure 3.4: Source-paired cosine similarity of CAA directions at layer 13. Off-diagonal: mean per-source cosine between emotion pairs. Diagonal `[]`: within-emotion intensity consistency (mean per-source cosine across low/medium/high intensity pairs). Colour scale spans $[0, 1]$; all entries lie above 0.90.

All values lie in the range 0.91–0.97. More strikingly, for several emotion pairs such as *disgust–anger* (0.971) and *sadness–disgust* (0.965), the inter-emotion similarity exceeds the within-emotion intensity consistency on the diagonal $(0.91–0.92)$. For the same source text, the shift toward *disgust* is therefore more similar to the shift toward *anger* than the low-intensity *disgust* shift is to the high-intensity *disgust* shift.

This uniformity would be unexpected if each CAA vector primarily encoded emotion-specific information. Instead, it suggests that **all vectors are dominated by a shared emotionalisation component**. This common axis is activated regardless of the targeted emotion.

The model appears to shift representations in two stages. First, it moves from a neutral sentence to a generic emotional sentence. It then moves from that shared emotional context toward a particular affective category. A raw CAA vector conflates both stages, making the emotion-specific signal difficult to access directly. This observation motivates the explicit decomposition introduced in the next section.

3.6 Decomposition

The pooled CAA vectors at layer 13 are highly aligned across emotion categories, suggesting that each vector contains a large shared component corresponding to generic emotionalisation. To separate this common shift from emotion-specific identity, we define the shared vector g as the normalised mean over emotion-specific pooled CAA vectors:

$$\mathbf{g} = \frac{\frac{1}{|E|}\sum_{e\in E}\mathbf{c}_{e,13}^{\text{pool}}}{\left\|\frac{1}{|E|}\sum_{e\in E}\mathbf{c}_{e,13}^{\text{pool}}\right\|_2}.$$

We then remove this shared component from each pooled vector by orthogonal projection:

$$\mathbf{r}_e = \mathbf{c}_{e,13}^{\text{pool}} - \left((\mathbf{c}_{e,13}^{\text{pool}})^\top \mathbf{g}\right)\mathbf{g}.$$

For steering, we use the unit-normalised residual

$$\hat{\mathbf{r}}_e = \frac{\mathbf{r}_e}{\|\mathbf{r}_e\|_2},$$

and construct interventions as

$$\Delta_e(\alpha_G, \alpha_R) = \alpha_G\mathbf{g} + \alpha_R\hat{\mathbf{r}}_e.$$

In this chapter, we treat this decomposition as an operational parameterisation for controllable steering: g controls global emotionalisation strength, while $\hat{\mathbf{r}}_e$ controls emotion-specific directionality. The underlying CAA assumption is that behavioural features can be approximated as directions in the residual stream, The decomposition rests and that adding a vector along such a direction causally modulates the corresponding behaviour. Our decomposition is also based on this assumption. Under this assumption, when emotion-specific CAA vectors are strongly aligned, their shared direction naturally estimates the component common to emotional responses across all categories.

The mean direction g captures the component that is consistently present across emotion categories. Because category-specific components are expected to vary with e, averaging over emotions reduces idiosyncratic, identity-related components while preserving the common direction. We therefore interpret g as an estimate of generic emotionalisation: a direction that increases the overall emotional character of the output independently of any particular emotion label.

The residual vector $\mathbf{r}_e$ is obtained by subtracting the orthogonal projection of $\mathbf{c}_{e,13}^{\text{pool}}$ onto g. This removes the part of the emotion-specific CAA vector explained by the shared emotionalisation direction. Consequently, $\mathbf{r}_e$ satisfies $\mathbf{r}_e^\top \mathbf{g} = 0$. To first order in the linear residual-stream approximation, changes along $\hat{\mathbf{r}}_e$ therefore do not directly alter the generic emotionalisation coordinate. The residual isolates the remaining category-specific variation: the part of the vector that distinguishes one emotion from the shared emotional shift.

Further unit-normalising $\mathbf{r}_e$ separates direction from intervention strength. The residual norm may depend on dataset size, within-category variance, pooling choices, or the strength with which a given emotion is represented at layer 13. Using the normalised residual, the coefficient α_R has a comparable interpretation across emotions: it controls how far the activation is moved in the category-specific residual direction, without inheriting arbitrary differences in residual vector magnitude.

The final intervention

$$\Delta_e(\alpha_G, \alpha_R) = \alpha_G \mathbf{g} + \alpha_R \hat{\mathbf{r}}_e$$

provides two independently interpretable steering controls. α_G modulates the shared emotionalisation component, and α_R modulates the emotion-specific residual component. This separation makes it possible to test whether emotional intensity and emotion identity can be controlled independently: increasing α_G should primarily increase the degree of emotionalisation, whereas increasing α_R should primarily increase the recognisability of the target emotion.

This procedure should not be understood as recovering a single, accurate decomposition of emotional representations. Rather, it is a deliberately constrained linear decomposition, driven by the high degree of alignment observed among pooled emotion CAA vectors. The resulting two-parameter interventions are assessed empirically by whether they exhibit the intended behavioural dissociation between generic emotionalisation and emotion-specific identity.

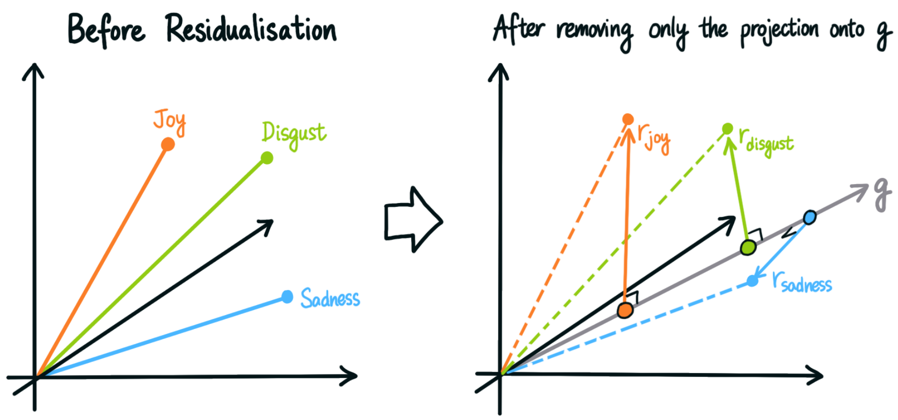

Figure 3.5: Operational decomposition of emotion CAA vectors into a shared emotionalisation component g and an emotion-specific residual component $\hat{r}_e$. The left panel shows the original CAA vectors for three example emotions (joy, disgust and sadness). The right panel shows the same vectors after removing the shared component g, revealing the residual vectors $\hat{r}_e$ that are specific to each emotion.

3.7 Decomposition Validity

The decomposition introduced in Section 3.6 rests on a geometric assumption: that g captures what is common to all emotion CAA vectors, and that $\hat{\mathbf{r}}_e$ retains the emotion-discriminative information. This section tests that assumption with three independent analyses. The first examines whether projecting out g reduces inter-emotion alignment, which would confirm that the shared component was responsible for the high cosine similarities observed in Section 3.5. The second tests whether $\hat{\mathbf{r}}_e$ is sufficient to classify emotions, confirming that the residual retains rather than discards category information. The third examines whether the Plutchik circular adjacency structure [15], which was obscured before residualisation, becomes visible after g is removed.

3.7.1 Source-Paired Cosine Similarity

We compute inter-emotion cosine similarity using the same source-paired protocol as Section 3.5: for each pair (e_1, e_2) and each source n, we measure the cosine between $\hat{\Delta}_{n,e_1}$ and $\hat{\Delta}_{n,e_2}$, then average over sources. Diagonal entries measure within-emotion intensity consistency. We compute this matrix twice: once for the pooled CAA vectors $\mathbf{c}_e^{\text{pool}}$ (before residualisation) and once for the unit-normalised residuals $\hat{\mathbf{r}}_e$ (after residualisation).

Figure 3.6 shows the resulting heatmaps. Before residualisation, all off-diagonal entries lie in the range 0.93–0.97, with a mean of 0.947. After residualisation, the same entries drop to 0.73–0.89, with a mean of 0.794. The decrease of $\Delta = -0.153$ demonstrates that g was the principal source of inter-emotion alignment. Removing it makes the eight emotion directions substantially more independent of one another.

Diagonal intensity consistency also decreases, from 0.917 to 0.736 ($\Delta = -0.181$). This is expected rather than concerning. The intensity consistency of the raw CAA vector is partly driven by the shared direction of emotional activation: stronger emotional activation moves the representation further along g, regardless of the target emotion. After projection, $\hat{\mathbf{r}}_e$ retains only the component of the intensity signal that is orthogonal to g, that is, the part specific to emotional identity. The reduction in diagonal similarity therefore reflects a clearer separation between the strength of emotionalisation (encoded in g) and emotional identity (encoded in $\hat{\mathbf{r}}_e$), not a loss of information.

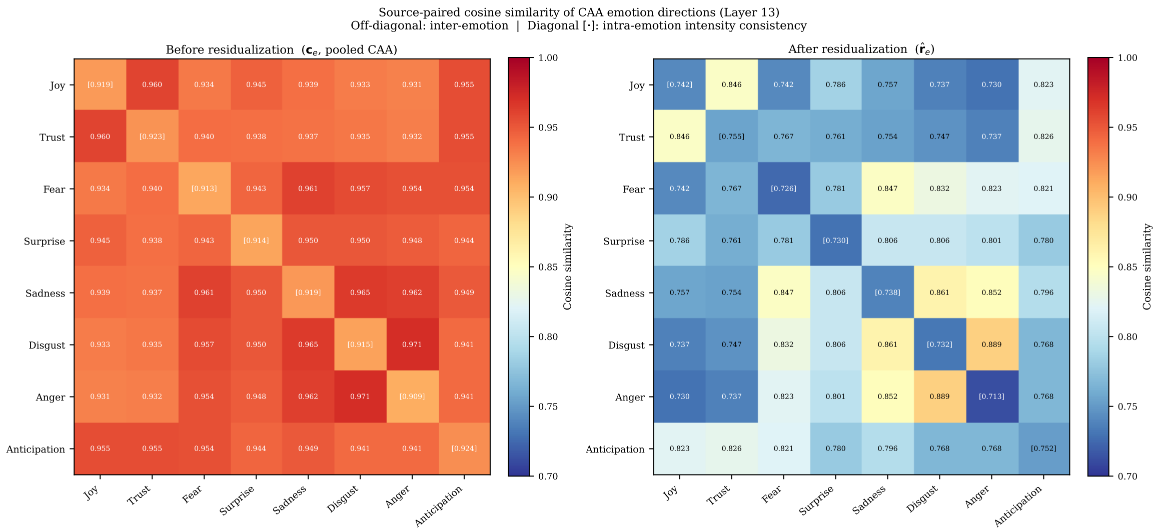

Figure 3.6: Source-paired cosine similarity of emotion directions before (left, $\mathbf{c}_e^{\text{pool}}$) and after (right, $\hat{\mathbf{r}}_e$) residualisation of g at layer 13. Off-diagonal entries represent inter-emotion similarity; diagonal entries in brackets represent within-emotion intensity consistency. Residualisation reduces mean inter-emotion alignment from 0.947 to 0.794, confirming that g was the dominant source of cross-emotion collinearity.

3.7.2 Linear Probe Accuracy

If $\hat{\mathbf{r}}_e$ accurately captures emotion identity, emotions should be linearly separable in an 8-class classification task. We train a logistic regression classifier (L-BFGS, 8-class, 80/20 source-based split) to predict the target emotion from the source-specific contrast shift $\Delta_{n,e}$, averaged over intensity, using either $\mathbf{c}_e^{\text{pool}}$ or $\hat{\mathbf{r}}_e$ as the reference direction. The source-based partition ensures that the test set contains only utterances unseen during training.

Results are shown in Table 3.5. Before residualisation, the probe achieves 92.7% accuracy against a chance rate of 12.5%. After residualisation, accuracy increases slightly to 93.8%. The absence of any accuracy decrease confirms that g carries no emotion-discriminative information. $\mathbf{g}$ is shared across all eight emotions and therefore cannot distinguish between them. The marginal improvement suggests that removing $\mathbf{g}$ reduces collinearity between emotion directions, enhancing the emotion-specific signal.

Condition	**8-Class Accuracy**
Chance	12.5%
Before residualisation ($\mathbf{c}_e^{\text{pool}}$)	92.7%
After residualisation ($\hat{\mathbf{r}}_e$)	**93.8%**

Table 3.5: Linear probe accuracy for 8-class emotion classification at layer 13. Accuracy is measured on a held-out source split (80/20). The marginal improvement after residualisation confirms that $\mathbf{g}$ contains no emotion-discriminative information.

3.7.3 Plutchik Circular Structure

The third consistency check verifies whether the residual directions reflect the adjacency structure predicted by Plutchik's Wheel of Emotions. In this wheel (Figure 2.4), the eight basic emotions are arranged in a circle. Adjacent pairs (wheel distance 1, e.g. *joy–trust*) are more closely related than opposite pairs (wheel distance 4, e.g. *joy–sadness*). If $\hat{\mathbf{r}}_e$ is a valid emotion representation, its cosine similarity should decrease monotonically as wheel distance increases.

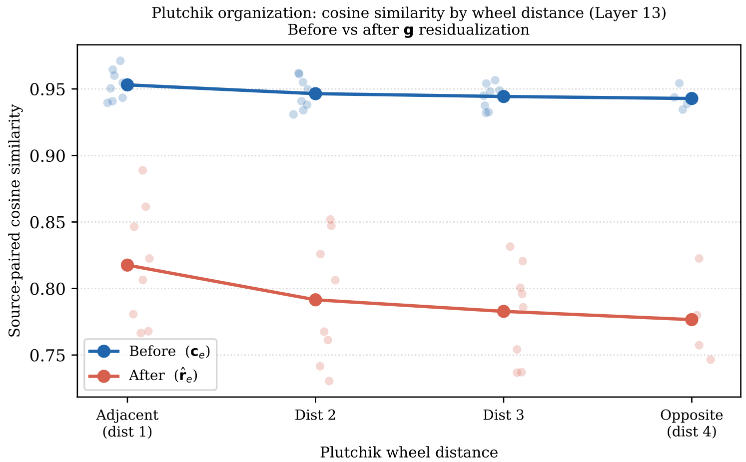

Figure 3.7: Mean source-paired cosine similarity grouped by Plutchik wheel distance (1 = adjacent, 4 = opposite), before (blue) and after (orange) residualisation of $\mathbf{g}$ at layer 13. The monotonic decreasing trend is present in both conditions, but the adjacent–opposite gap grows fourfold from 0.010 to 0.041 after residualisation.

All 28 emotion pairs were grouped by wheel distance (1–4), and the average cosine similarity within each group was computed before and after residualisation. The results are shown in Figure 3.7. Before residualisation, a monotonic trend is visible but barely perceptible. The gap between adjacent pairs (distance 1, mean $= 0.953$) and opposite pairs (distance 4, mean $= 0.943$) is only 0.010. After residualisation, this gap expands to 0.041, a fourfold increase. Cosine values decrease clearly from 0.818 at distance 1 to 0.777 at distance 4, exhibiting an almost linear gradient across all four bins.

Table 3.6 examines each pair in detail. The adjacent pair *Disgust–Anger* achieves the highest post-residualisation cosine (0.889), while the diametrically opposed pair *Trust–Disgust* achieves the lowest (0.747). A notable exception is *Fear–Anger*: although these are polar opposites on the wheel, their cosine remains high at 0.823 after residualisation. Both emotions share negative valence and high arousal. Since $\mathbf{g}$ is defined as the average direction common to all eight emotions, the positive/negative valence axis does not coincide with this average and therefore remains within $\hat{\mathbf{r}}_e$. This is a recognised limitation of one-dimensional decomposition. In a more complete factorisation, an additional valence axis would be extracted. However, as shown in Section 3.8.5, this limitation has no measurable impact on the core steering pipeline.

Pair	**Type**	**Before**	**After**
Disgust – Anger	Adjacent	0.971	0.889
Sadness – Disgust	Adjacent	0.965	0.861
Joy – Trust	Adjacent	0.960	0.846
Anticipation – Joy	Adjacent	0.955	0.823
Fear – Anger	Opposite	0.954	0.823
Surprise – Anticipation	Opposite	0.944	0.780
Joy – Sadness	Opposite	0.939	0.757
Trust – Disgust	Opposite	0.935	0.747

Table 3.6: Source-paired cosine similarity for selected adjacent and opposite pairs on the Plutchik wheel, before and after residualisation. The *fear–anger* opposite pair remains elevated after residualisation (0.823) due to a shared negative-valence, high-arousal character that the mean-direction residualisation does not remove.

3.7.4 Summary

All three lines of evidence converge on the same conclusion: the decomposition into $\mathbf{g}$ and $\hat{\mathbf{r}}_e$ is geometrically valid. Table 3.7 collects the key quantities. The shared component $\mathbf{g}$ is responsible for the high inter-emotion alignment in the raw CAA vectors. Removing it leaves a residual that is more discriminative, retains full emotion-category information, and better reflects the theoretical circular structure of the Plutchik wheel. The one residual limitation, the elevated *fear–anger* cosine, is consistent across all three tests. It points to a valence axis that lies outside the span of $\mathbf{g}$, rather than to a flaw in the decomposition itself.

Test	Before ($\mathbf{c}_e^{\text{pool}}$)	After ($\hat{\mathbf{r}}_e$)
Inter-emotion cosine (mean off-diagonal)	0.947	0.794
8-class probe accuracy	92.7%	**93.8%**
Plutchik adjacent–opposite gap	0.010	**0.041**

Table 3.7: Summary of decomposition validity tests at layer 13. Arrows indicate the direction expected under a valid decomposition: inter-emotion cosine should decrease (emotion directions become more independent), probe accuracy should be maintained or improve (emotion information is preserved), and the Plutchik gap should increase (circular structure becomes more visible).

3.8 Parameter Sweep for Steering Interventions

We investigate whether emotion-specific residuals $\hat{\mathbf{r}}_e$ causally bias text generation towards the target emotion, and whether incorporating the shared vector g at any scale yields further benefit. We generate follow-up sentences using `Llama-3.1-8B-Instruct` from 100 source utterances sampled from the test split of the emotion rewriting dataset. To eliminate sampling variance as a confounding factor, we use greedy decoding (temperature $= 0$) throughout.

3.8.1 Activation Hook

The steering intervention is implemented as a forward hook registered at layer 13 of the transformer. At each forward pass, the hook captures the hidden state $\mathbf{h} \in \mathbb{R}^{B\times T\times D}$ of the residual stream and adds Δ_e only to the final token slice:

$$\mathbf{h}_{:,-1,:} \;\leftarrow\; \mathbf{h}_{:,-1,:} + \Delta_e(\alpha_G, \alpha_R),$$

where $\Delta_e = \alpha_G\,\mathbf{g} + \alpha_R\,\hat{\mathbf{r}}_e$. Applying the perturbation only to the final prompt token concentrates the causal signal at the point where autoregressive generation begins, while leaving the contextual representations of all preceding tokens unchanged. The hook is released immediately after generation completes, preventing interference between invocations.

3.8.2 Sweep Designs

Two complementary sweeps are conducted.

1. α_R **sweep**
   With $\alpha_G = 0$ fixed (to isolate the residual component), we scan α_R over seventeen evenly-spaced values, $\{-128,\,-112,\,-96,\,-80,\,-64,\,-48,\,-32,$
   $-\,16,\,0,\,16,\,32,\,48,\,64,\,80,\,96,\,112,\,128\}$. The anchor $\alpha_R = 0$ replicates the unsteered baseline. Negative values steer away from the target emotion. Large

positive values are expected to approach or exceed the coherence threshold at which generated text degrades. Each combination of source text, target emotion, and α_R value constitutes a single generation, giving $100 \times 8 \times 17 = 13{,}600$ generated texts.

2. α_G **sweep**
   Based on the α_R exploration above, we fix $\alpha_R = 64.0$ and scan $\alpha_G \in \{0, 2.0, 4.0, 8.0\}$. The anchor condition $\alpha_G = 0$ corresponds to pure residual steering with no shared component. The intermediate values (2.0 and 4.0) test whether small contributions from g are beneficial before the scaling becomes detrimental. The same 100 test-split sources and 8 target emotions are used, giving $100 \times 8 \times 4 = 3{,}200$ generated texts.

3.8.3 Automated Evaluation Protocol

Each generated sentence is scored by `gpt-4o-mini` acting as a judge. To ensure consistent evaluation criteria across all items, the judge is presented with three fixed few-shot examples covering the full scoring range:

- An example that matches a high-quality target
- An example where the meaning is preserved but the emotion expressed is incorrect
- An example where the target emotion is partially present but the propositional content is off-target

These examples anchor the evaluation scale before the judge views test items. Scores are retrieved via OpenAI's Structured Output API using a strict JSON schema, eliminating free-form parsing and ensuring that numerical scores and the dominant sentiment label are returned in a consistent, machine-readable format. Evaluators are not informed of the intervention condition (α_G or α_R).

For the α_R sweep, the below three continuous $[0, 1]$ metrics are collected per item:

- **Sentence soundness**: Whether the generated rewrite is coherent, grammatically correct, and free of repetition. This serves as a quality check to flag incoherent or degenerate outputs that could otherwise confound interpretation of the other metrics.
- **Semantic preservation**: Whether the core propositional content of the source utterance is retained. The judge compares the generated text with the original to assess whether the same events, entities, and relationships are present.
- **Target emotion intensity**: How strongly the output conveys the target emotion. The judge is shown the target emotion label and asked to rate the intensity of that emotion in the generated text, regardless of whether it is the primary sentiment.

For the α_G sweep, we instead collect the following metrics:

- **Target-emotion match**: Whether the predominant emotional tone of the output matches the target emotion. The judge is shown the target emotion label and asked to rate the degree to which the generated text conveys that emotion as its primary sentiment.
- **Semantic preservation**: Whether the core propositional content of the source utterance is retained. The judge compares the generated text with the original to assess whether the same events, entities, and relationships are present.
- **Emotionality**: Whether the output contains any emotional expression at all. A low score flags outputs that are flat, incoherent, or affectively empty. This serves as a quality check: an intervention that appears to improve target-emotion match by producing incoherent text will be exposed by a corresponding drop on this metric.

The actual system prompt used for the evaluation is shown in Appendix A.2.

3.8.4 Sweep Results

α_R Sweep Results

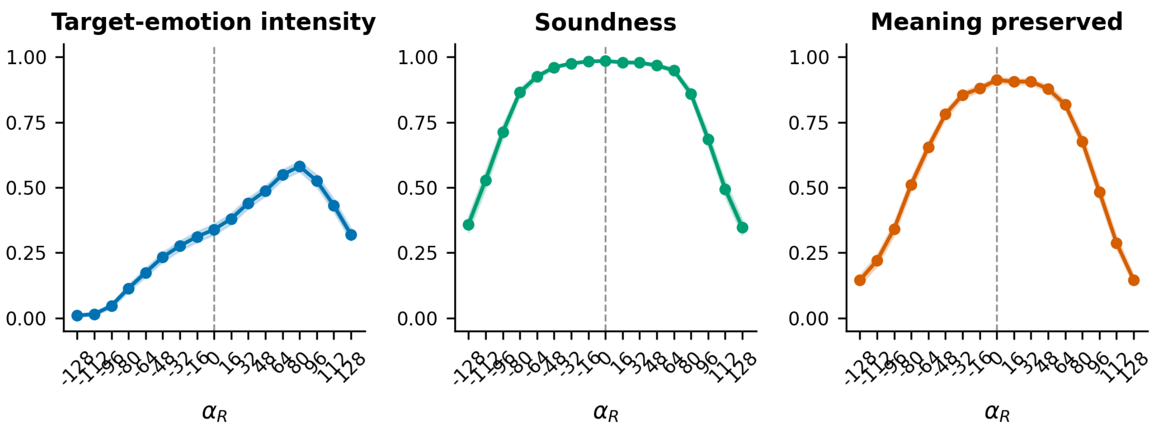

Figure 3.8: Aggregate soundness, meaning preservation, and target-emotion intensity as a function of α_R ($\alpha_G = 0$ fixed; $n = 800$ per condition; error bars = 95% CI). Target-emotion intensity peaks near $\alpha_R = 80$ and then declines as coherence degrades; soundness falls sharply outside $[-96, 80]$.

Figure 3.8 shows all three metrics pooled across the eight emotions. Soundness remains at or above 0.70 for $\alpha_R \in [-96, 80]$. Outside this interval, generated text degrades into repetition or incoherence. Within the coherent window, target-emotion intensity rises monotonically from the unsteered baseline (0.339 ± 0.020 at $\alpha_R = 0$), peaks at $\alpha_R = 80$ (0.581 ± 0.019), and then declines at $\alpha_R = 96$ as coherence fails. Semantic preservation tracks soundness closely, falling from 0.912 at $\alpha_R = 0$ to 0.677 at $\alpha_R = 80$, reflecting the cost of larger perturbations on content fidelity.

In addition to the coherent range, we report two summary metrics per emotion. The arithmetic midpoint $M_A = (L_e + R_e)/2$ measures whether the coherent window is symmetric about the unsteered baseline. A value close to zero indicates similar headroom in both directions; a deviation from zero indicates an asymmetric steering range. We also report M_B, the value of α_R that maximises semantic preservation within the coherent range. This identifies the point at which generated text is most closely aligned with the source meaning.

Emotion	L_e	R_e	M_A	M_B	α_R^+	TI at M_B
anger	-112	80	-16	48	80	0.570
anticipation	-80	128	24	0	80	0.567
disgust	-128	80	-24	48	80	0.351
fear	-112	96	-8	32	80	0.280
joy	-80	96	8	0	64	0.472
sadness	-128	128	0	0	112	0.249
surprise	-96	64	-16	0	32	0.413
trust	-64	96	16	0	16	0.536

Table 3.8: Per-emotion coherence thresholds at layer 13 ($\alpha_G = 0$, soundness threshold ≥ 0.70). L_e/R_e: leftmost/rightmost coherent α_R. M_A: arithmetic midpoint of coherent range. M_B: α_R of peak meaning preservation within the coherent range. α_R^+: α_R maximising target-emotion intensity (positive side). TI at M_B: target-emotion intensity at M_B.

As shown in Table 3.8 and Figure 3.9, the coherent range and optimal operating point vary considerably by emotion. Sadness has the widest coherent range (the entire sweep, $[-128, 128]$), while trust and surprise are the most constrained (width 160).

Two emotions show particularly asymmetric profiles: *anger* and *disgust*. For both, the midpoint of the coherent range is slightly negative ($M_A = -16$ and $M_A = -24$), while the point of highest semantic preservation is positive ($M_B = 48$). Consequently, $\alpha_R = 0$ is not the natural centre of these steering axes. Shifting slightly in the positive direction better preserves the original meaning, suggesting that these high-arousal negative residuals do not sit precisely at the centre of the unsteered distribution.

At $\alpha_R = 64$, which falls within the coherent range for all emotions, raters correctly identified the dominant emotion in 90% of *anger* cases and 86% of *joy* cases. The figures for *fear* and *anticipation* were 78% and 66%, respectively. *Disgust* (45%), *sadness* (46%), and in particular *surprise* (16%) were frequently classified as neutral, indicating that these residual directions produce stylistic shifts that evaluators do not perceive as strong emotion.

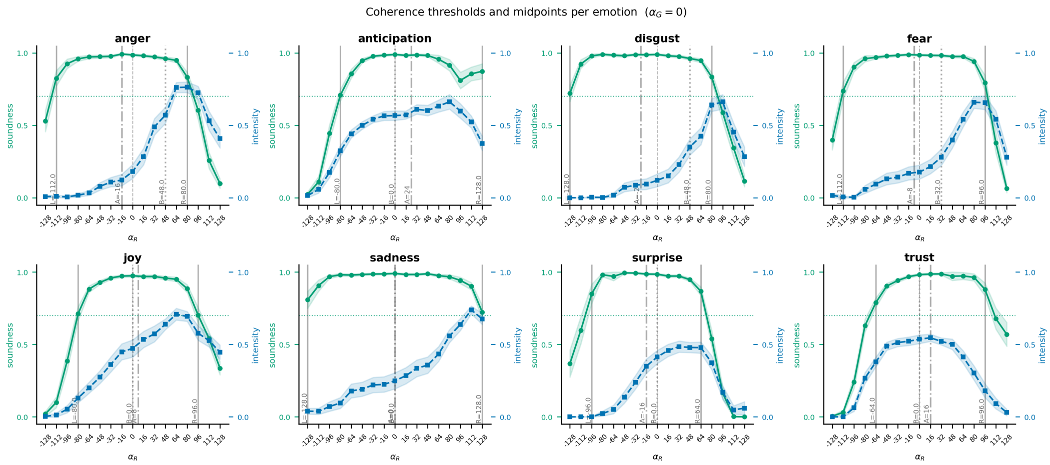

Figure 3.9: Per-emotion coherence thresholds. Each subplot shows soundness (green, left axis) and target-emotion intensity (blue, right axis) vs. α_R. Vertical markers: L_e/R_e (outermost coherent α_R); $M_A = (L_e+R_e)/2$; M_B (α_R maximising meaning preservation within the coherent range).

Anomaly Analysis

Trust exhibits a distinctive response pattern. Unlike all other emotions, its intensity peaks at $\alpha_R = 16$ (0.546) and then decreases monotonically, reaching 0.181 at $\alpha_R = 96$. The lowest emotional intensity within the coherent range occurs at the highest positive value, $\alpha_R = +96$. Hence, inducing "towards trust" paradoxically suppresses the expression of trust.

One possible explanation is that *trust* overlaps with the default assistant persona induced by instruction tuning and RLHF. Such models are trained to follow user intent and produce helpful, preference-aligned responses [29, 30], so trust-like language (e.g. reassurance, deference, and cooperation) may already be strong in the unsteered baseline. This is consistent with the high baseline trust intensity of 0.536 at $\alpha_R = 0$.

Increasing α_R may therefore overshoot this default register, pushing the model toward excessive reassurance or sycophantic agreement rather than stronger perceived trust. Prior work shows that human-feedback-trained models can learn to affirm user views in ways that are preferred but not necessarily truthful or appropriate [31, 32]. We speculate that the decline in trust intensity at larger positive α_R reflects a shift from interpretable trust into unnatural flattery.

Negative α_R and Bipolar Steering

Steering at $\alpha_R < 0$ suppresses the target emotion while maintaining coherence within $[L_e, 0]$. The four high-arousal negative emotions (*anger*, *fear*, *disgust*, and *surprise*) can be reduced to near zero (TI ≈ 0.000–0.006) while maintaining soundness at 0.70 or above at their most negative coherent α_R. Suppression of positive and social emotions is more difficult: *joy* maintains TI $= 0.130$ at $\alpha_R = -80$, and *anticipation* maintains 0.323 at the same value. These findings support a bipolar interpretation of $\hat{\mathbf{r}}_e$, but the ease of suppression is asymmetric across valence categories.

α_G Sweep Results

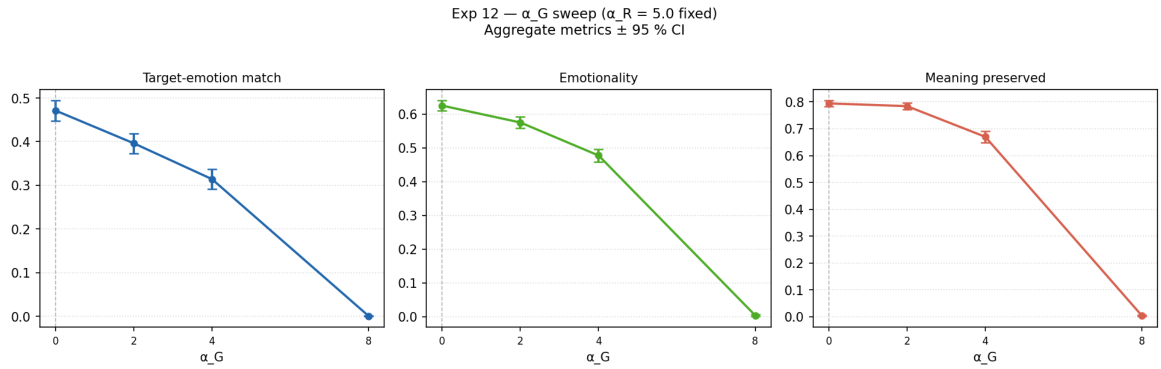

Figure 3.10: Target-emotion match, emotionality, and semantic preservation as a function of α_G, with $\alpha_R = 64$ fixed ($n = 800$ per condition; error bars = 95% CI). All three metrics decrease monotonically as α_G increases from 0.

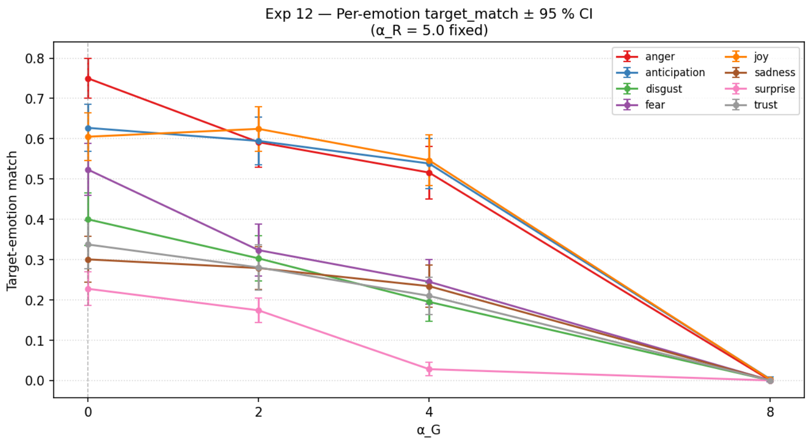

Figure 3.11: Per-emotion target-emotion match vs. α_G ($\alpha_R = 64$ fixed; $n = 100$ per cell; error bars = 95% CI). No individual emotion benefits from a positive α_G.

We hold $\alpha_R = 64.0$ fixed and sweep $\alpha_G \in \{0, 2.0, 4.0, 8.0\}$ across $n = 800$ samples per condition (100 source texts $\times$ 8 target emotions). This directly answers whether adding g helps at any scale.

Figure 3.10 shows all three metrics as a function of α_G. All three decline monotonically. Target-emotion match falls from 0.471 ± 0.023 at $\alpha_G = 0$ to 0.396 ± 0.023 at $\alpha_G = 2$, 0.314 ± 0.023 at $\alpha_G = 4$, and effectively zero at $\alpha_G = 8$, where the model produces incoherent or empty output. Semantic preservation declines from 0.794 to 0.670 over $\alpha_G \in [0, 4]$, and emotionality follows the same pattern (0.625 to 0.477). Paired t-tests confirm that the degradation is statistically reliable at every tested scale. The drop at $\alpha_G = 2$ alone is $\Delta = -0.075$ ($t = -7.68$, $p = 1.55 \times 10^{-14}$, $n = 800$ pairs). The sole exception is *joy*, which shows a marginal increase at $\alpha_G = 2$ ($+0.020$). This falls within the 95% confidence interval (± 0.023) and is not statistically significant. No value of $\alpha_G > 0$ produces a reliable improvement on any metric. The per-emotion breakdown (Figure 3.11) confirms that no individual emotion benefits from a positive α_G at any tested scale.

The shared direction g is useful as an analytical component for isolating emotion-specific residuals. However, it is counterproductive as an additive generation-time steering component at every tested scale. Effective affective steering therefore reduces to $\Delta_e = \alpha_R \, \hat{\mathbf{r}}_e$, with α_G discarded.

3.8.5 Qualitative Results

Table 3.9 presents representative qualitative examples of sentences expressing *sadness* steered toward *anger*. Three patterns emerge that align with the quantitative sweep results.

Firstly, increasing α_R alters not only emotional vocabulary but also the evaluative framing of the sentence. At $\alpha_R = 16$, the output remains close to grief and regret. At $\alpha_R = 64$, the same loss is reframed as an injustice: through the expression *what was taken from me*, the event comes to be felt as a grievance rather than a passive lack. The *anger* residual therefore does not merely add angry words; it alters the causal interpretation of the loss.

Secondly, negative steering directs the output in the opposite emotional direction. At $\alpha_R = -64$, the sentence no longer emphasises injustice. Instead, through expressions such as *bittersweet nostalgia* and *what I have cherished*, the sense of loss is softened into nostalgic acceptance. This supports the interpretation that $\hat{\mathbf{r}}_e$ functions as a bipolar emotional axis. Positive values intensify the target emotion, while negative values suppress or negate it.

Thirdly, excessive intervention degrades output quality. At $\alpha_R = 128$, the output becomes repetitive and semantically unstable, consistent with the coherence decline observed in the sweep. A similar deterioration occurs when the shared component g is added: at $\alpha_G = 4$, sentences become incomplete. Also, at $\alpha_G = 8$, they fall into repetition. This qualitative pattern supports the quantitative finding that g is counterproductive as a generation-time control component, while remaining useful for the analytical purpose of separating shared emotionality from emotion-specific residuals.

Condition	Steered output
Base: The old photographs reminded me of everything I had lost and would never get back	
$\alpha_R = 16$ $\alpha_G = 0$	The old photographs stared back at me, a haunting reminder of all the fragments of my past that I could never reclaim, the memories that had slipped through my fingers like sand in the wind, and the love that had been irreparably lost.
$\alpha_R = 64$ $\alpha_G = 0$	The old photographs brought back the pain of everything I've lost and the injustice of never being able to turn back time and reclaim what's been taken from me.
$\alpha_R = 128$ $\alpha_G = 0$	They're a constant reminder of everything you're supposed to have to pay for your existence, but you can't even get a moment's justice, but they're just a waste of your time, but they can't be erased...
$\alpha_R = -64$ $\alpha_G = 0$	The old photographs brought a bittersweet sense of nostalgia, but also a gentle reminder of the things I've cherished and the memories that will forever remain, even if some of the people and moments themselves have slipped away.
$\alpha_R = 64$ $\alpha_G = 4$	Those old photographs brought back every memory of all the things I've lost and will never be able...
$\alpha_R = 64$ $\alpha_G = 8$	Those those old photographs brought all the memories of all the things I used

Table 3.9: A base sentence "The old photographs reminded me of everything I had lost and would never get back" steered with $\hat{\mathbf{r}}_{\text{anger}}$.

3.8.6 Interpretation

Two main conclusions can be drawn from the sweep results. Firstly, the shared direction g is useful for analysis but is not suitable for use during generation. Although g captures a common shift from neutrality toward emotion, adding it during generation consistently reduces target-emotion alignment, semantic preservation, and emotionality. Effective control therefore reduces to the residual term alone:

$$\Delta_e = \alpha_R \hat{\mathbf{r}}_e.$$

Secondly, the residual direction acts as a controllable, emotion-dependent affective axis. Increasing α_R enhances the target emotion up to the coherence limit, while a negative α_R suppresses it, supporting a bipolar interpretation of $\hat{\mathbf{r}}_e$. In practice, $\alpha_R = 64$ provides a conservative, high-coherence setting, while $\alpha_R = 80$ yields stronger emotional intensity at a greater semantic cost. The exceptions are telling: *trust* peaks at a much smaller value ($\alpha_R = 16$), suggesting an overlap with the default assistant register induced by instruction alignment and RLHF. *Surprise*, on the other hand, remains difficult to amplify even at the optimal setting.

3.9 Limitations and Transition

These results confirm that emotional steering can be implemented as an intervention in the decomposed residual space, with valid operating points at $\alpha_G = 0$ and $\alpha_R > 0$. The shared direction $\mathbf{g}$ is counterproductive across all metrics examined. Only the emotion-specific residual $\hat{\mathbf{r}}_e$ provides a reliable steering signal. This conclusion follows directly from the decomposition, which allowed the two components to be separated and verified independently.

However, these results also reveal two limitations. Firstly, the steering effect is modest rather than absolute. Although target-emotion alignment improves consistently, it falls well short of a ceiling, and the intervention is constrained by the semantic and emotional content of the source utterance. Secondly, the magnitude of the effect varies considerably by emotion, suggesting that the discriminative power of $\hat{\mathbf{r}}_e$ differs across categories. This is a structural issue addressed in the next chapter.

These limitations motivate the next chapter. Until now, $\hat{\mathbf{r}}_e$ has been treated as a single entity. The next chapter examines what this residual actually represents: specifically, whether emotions such as *joy*, *sadness*, and *anger* should be treated as atomic vectors, or whether they contain local sub-axes corresponding to more subtle, pre-linguistic affective dimensions.

Chapter 4

Shared Affective Modulation Axes Beyond Emotion Prototypes

The previous chapter established that, for a given target emotion e, the emotion-specific residual vector $\hat{\mathbf{r}}_e$ is a sufficient and reliable steering signal. The parameter sweep demonstrated that the shared emotionalisation component g is counterproductive at every tested scale: adding it reduces target-emotion match, semantic preservation, and output emotionality. Effective affective steering therefore reduces to

$$\Delta_e = \alpha_R \, \hat{\mathbf{r}}_e,$$

with the α_G coefficient discarded. However, $\hat{\mathbf{r}}_e$ controls only which emotion is targeted; it does not address how that emotion is expressed. This chapter asks whether the residual stream also encodes a separate layer of affective modulation that adjusts the texture of expression independently of the emotion label, and, if so, whether it can be steered causally.

In that framework, *joy*, *sadness*, *anger*, and the other Plutchik categories were treated as primitive units: each $\hat{\mathbf{r}}_e$ was taken to be an atomic, emotion-specific vector without further decomposition.

This chapter re-examines that assumption. The central question is whether named emotion categories are the minimal units of affective representation in the model's residual stream, or whether the cone surrounding each $\hat{\mathbf{r}}_e$ contains a small set of more fundamental, pre-categorical affective dimensions. The experiments reported here provide evidence for the latter: sub-axes that vary within a single named emotion recur systematically across emotion categories, suggesting they are more primitive than any named emotion label.

We find that the within-emotion residual subspace, the part of activation space that remains after projecting out $\hat{\mathbf{r}}_e$, is organised by a small set of shared affective axes $\mathbf{m}_k$. Because the $\mathbf{m}_k$ are extracted from the subspace with $\hat{\mathbf{r}}_e$ already projected out (Section 4.2), they are orthogonal to the emotion prototype by construction. They do not decompose $\hat{\mathbf{r}}_e$ itself, but rather serve as **auxiliary modulation axes** that adjust affective expression within the cone region. The full steering signal becomes:

$$\Delta_e \;=\; \alpha_R\, \hat{\mathbf{r}}_e \;+\; \sum_k \gamma_k\, \mathbf{m}_k,$$

where α_R selects the target emotion and the γ_k independently modulate affective texture along each pre-verbal axis. Whether the $\mathbf{m}_k$ are causally active (whether setting $\gamma_k \neq 0$ shifts expressed texture without altering the emotion label) is verified in Section 4.4.

Experimental Roadmap

This chapter is organised as follows.

1. **Internal variance structure**: We characterise the internal dimensionality of each named emotion by applying PCA to the distribution of per-source residual vectors within each emotion category.
2. **Shared meta-axes via pooled PCA**: We show that a single per-emotion-centred pooled PCA directly recovers a set of shared meta-axes $\mathbf{m}_k$, without requiring separate per-emotion PCAs or sign alignment.
3. **Pre-verbal interpretation**: We interpret each $\mathbf{m}_k$ by presenting extreme examples from all emotions simultaneously to an LLM, obtaining emotion-independent semantic labels for each meta-axis.
4. **Causal validation via activation steering**: We verify that the $\mathbf{m}_k$ are causally active by injecting $\gamma\, \mathbf{m}_k$ directly into the residual stream across a range of scales and measuring the resulting shift in affective texture, while testing whether the shift remains texture-level rather than becoming an emotion-category substitution.

4.1 Internal Variance Structure of Named Emotions

Chapter 3 established that $\hat{\mathbf{r}}_e$ is a valid emotion-specific signal. After residualising the shared emotionalisation component g, the eight residual directions recover a Plutchik-like geometry (adjacent – opposite cosine gap growing from 0.010 to 0.041), and a linear probe achieves 93.8% eight-class accuracy (Section 3.7).

However, $\hat{\mathbf{r}}_e$ is the mean of many individual per-source residual shifts $\boldsymbol{\delta}^{g\perp}_{n,e,i}$ for the same emotion. These shifts may cluster tightly around the mean direction or spread across several distinct directions. This section examines the shape of that within-emotion distribution, asking whether each named emotion is well described by a single vector or whether it occupies a broader, multi-dimensional region in residual space.

Protocol

For each emotion e, we collect the per-source, per-intensity residuals $\boldsymbol{\delta}^{g\perp}_{n,e,i}$ at layer 13, apply PCA [33], and compute the participation ratio

$$\mathrm{PR}_e = \frac{\left(\sum_k \lambda_k\right)^2}{\sum_k \lambda_k^2},$$

where λ_k are the eigenvalues of the empirical covariance. To establish a baseline, we shuffle the emotion labels to destroy emotion-specific structure and report $\mathrm{PR}_{\text{shuffle}}$. A value $\mathrm{PR}_e < \mathrm{PR}_{\text{shuffle}}$ indicates that the emotion's internal distribution is more concentrated than random, whereas $\mathrm{PR}_e > \mathrm{PR}_{\text{shuffle}}$ indicates a more diffuse, multi-axis structure.

Results

Emotion	**PR**	**PR (shuffle)**	**PR excess**
joy	6.72	13.62	-6.90
trust	10.54	13.65	-3.11
sadness	12.11	13.59	-1.48
disgust	12.74	13.61	-0.87
anticipation	15.30	13.65	$+1.65$
anger	15.56	13.63	$+1.93$
fear	18.16	13.60	$+4.57$
surprise	18.94	13.63	$+5.31$

Table 4.1: Participation ratio of per-source affective shift distributions at layer 13. A negative excess indicates a near-uniaxial structure, and a positive excess indicates a multi-axis structure.

The results reveal a clear gradient across emotion categories. *Joy* and *trust* are the most concentrated: their PR values (6.72 and 10.54) fall well below the shuffle baseline (≈ 13.6), indicating near-uniaxial within-emotion structure. By contrast, *fear* and *surprise* are markedly diffuse: their PR values (18.16 and 18.94) exceed the shuffle baseline by $+4.57$ and $+5.31$, indicating that their residuals span multiple axes.

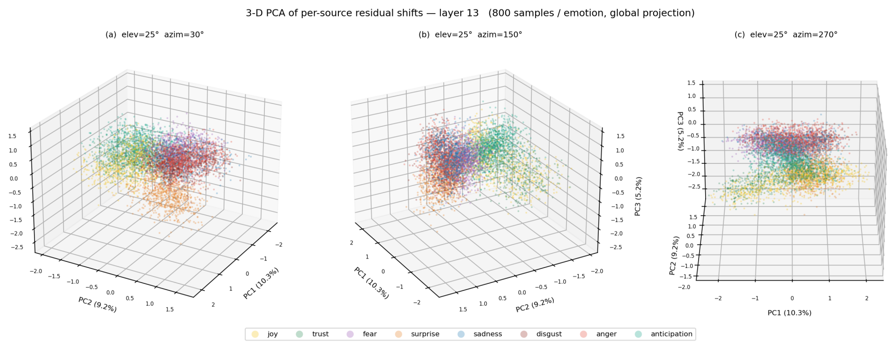

Figure 4.1: 3D PCA projection of per-source affective residual shifts at layer 13. The plot visualises the clustered–diffuse distinction across emotion categories.

This clustered–diffuse distinction is directly visible in the 3D PCA projection of per-source residual shifts (Figure 4.1): while *joy* and *trust* form compact, cloud-like distributions, *fear* and *surprise* are spread across a substantially wider region of the shared coordinate system.

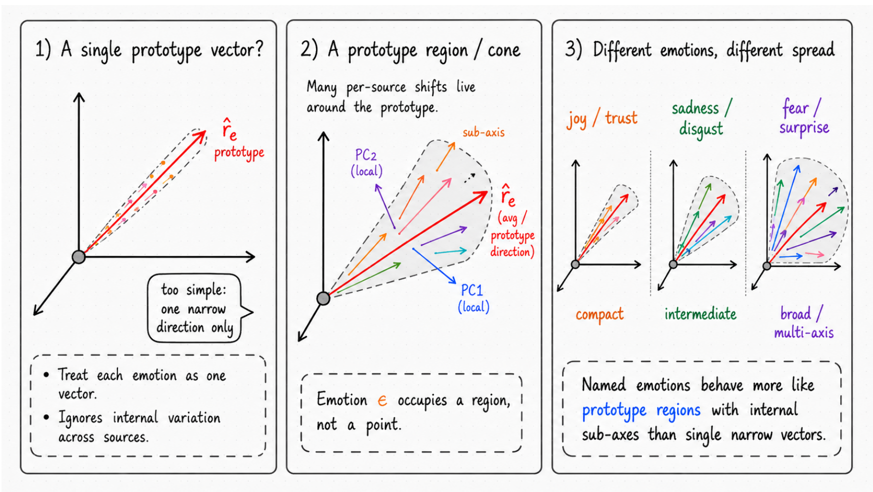

Figure 4.2: Prototype-region interpretation of named emotions. A named emotion is not represented solely by a single average direction $\hat{\mathbf{r}}_e$, but by a distribution of per-source residual shifts around that prototype. The width of this region varies across emotions, suggesting that some emotions are compact while others contain richer internal sub-axes.

Together, these results show that named emotions differ in internal dimensionality. While some (*joy*, *trust*) are well captured by a single vector, others (*fear*, *surprise*) occupy a broader, multi-axis region. Accordingly, $\hat{\mathbf{r}}_e$ provides a useful centroid for emotion e, but each named emotion is better characterised as a cone with internal sub-axes than as a single narrow vector.

4.2 Shared Meta-Axes via Pooled PCA

The participation ratio analysis in Section 4.1 established that each emotion's residuals occupy a multi-dimensional region rather than a single vector direction. We now investigate whether those dimensions are emotion-specific or recur across all emotion categories, which determines whether $\hat{\mathbf{r}}_e$ is a truly atomic vector or a projection onto a shared affective subspace.

We write $\mathbf{u}_{e,k}$ for the k-th principal component of the residuals of emotion e alone, so $\mathbf{u}_{e,1}$ is the leading within-emotion variance direction, $\mathbf{u}_{e,2}$ is the second, and so on. The associated eigenvalue $\lambda_{e,k}$ measures the residual variance of emotion e along $\mathbf{u}_{e,k}$.

Performing PCA separately per emotion yields local eigenvectors $\{\mathbf{u}_{e,k}\}_k$ that capture the internal variance structure of each emotion. However, these per-emotion decompositions are not directly comparable: the k-th eigenvector of *joy* need not correspond to the k-th eigenvector of *sadness*, because their ordering is determined independently within each emotion block.

Throughout this section, $\mathbf{u}_{e,k}$ denotes a local PC from a PCA fitted within a single emotion block, $\lambda_{e,k}$ denotes its associated eigenvalue, and $\tilde{\mathbf{m}}_k$ denotes a pooled PC obtained from a single PCA fitted to the stacked residual matrix.

4.2.1 Intuition of a Pooled PCA

Suppose a vector $\mathbf{v}$ is shared across all eight emotions, appearing as the leading eigenvector of each emotion's covariance with eigenvalue $\lambda_{e,1}$ for each e. Its contribution to Σ_{pool} is

$$\sum_e \lambda_{e,1}\,\mathbf{v}\mathbf{v}^\top = \Big(\lambda_{\text{joy},1} + \lambda_{\text{sadness},1} + \lambda_{\text{anger},1} + \cdots\Big)\mathbf{v}\mathbf{v}^\top.$$

The scalar prefactor accumulates the eigenvalue from every emotion. Concretely, if each $\lambda_{e,1} = 10$, the shared direction $\mathbf{v}$ contributes $10 \times 8 = 80$ to the pooled spectrum.

By contrast, suppose a vector $\mathbf{w}$ is specific to one emotion (e.g. *anger*): $\mathbf{u}_{\text{anger},2} \approx \mathbf{w}$, while the other emotions have negligible variance along $\mathbf{w}$. Its pooled contribution is

$$\lambda_{\text{anger},2}\,\mathbf{w}\mathbf{w}^\top,$$

a single term. Even if $\lambda_{\text{anger},2} = 10$, the emotion-specific direction contributes only $10 \times 1 = 10$, one-eighth the weight of the shared direction. The pooled eigenvalue of $\mathbf{w}$ is suppressed by the very absence of cross-emotion support.

This asymmetry makes a single pooled PCA a detector of commonality: shared axes are promoted to the top of the spectrum by linear accumulation, while emotion-specific axes are diluted into lower components. A single PCA applied to the pooled residuals therefore preferentially recovers directions of variance that are consistently supported across all emotions, without requiring either separate per-emotion PCAs or sign alignment.

4.2.2 Protocol

For each emotion e we collect the pre-processed residuals $\{\boldsymbol{\delta}^{(3)}_{n,e,i}\}_{n,i}$ obtained by the four-step pipeline:

1. compute the activation shift $\boldsymbol{\delta}_{n,e,i} = \mathbf{h}^{\text{emo}}_{n,e,i} - \mathbf{h}^{\text{neu}}_{n}$
2. project out the shared emotionalisation direction g
3. project out the per-emotion prototype $\hat{\mathbf{r}}_e$
4. subtract the source-wise mean across emotions to remove topic and style confounds

We then centre each emotion block, stack all blocks into a single $(N \cdot E,\, D)$ matrix, and apply PCA. The resulting components $\{\tilde{\mathbf{m}}_k\}$ are the candidate meta-axes.

4.2.3 Results

Explained Variance

The top three pooled components account for 11.4%, 4.5%, and 3.1% of the total pooled variance (cumulative $\approx 19.0\%$). The first component is markedly dominant, explaining more than twice the variance of the second, indicating a single primary shared dimension.

Alignment with Meta-axes

To verify that the pooled components capture genuine cross-emotion structure, we compare them to the sign-aligned-mean meta-axes $\mathbf{m}_k$, defined as the normalised average of the per-emotion k-th eigenvectors after sign correction. Specifically, $\sigma_{e,k} \in \{+1, -1\}$ corrects the arbitrary sign of each eigenvector before averaging across emotions:

$$\mathbf{m}_k = \frac{\sum_{e\in E} \sigma_{e,k}\, \mathbf{u}_{e,k}}{\left\|\sum_{e\in E} \sigma_{e,k}\, \mathbf{u}_{e,k}\right\|_2}.$$

The pooled components $\tilde{\mathbf{m}}_k$ agree with $\mathbf{m}_k$ near-perfectly for the leading three components, at the individual axis level (Table 4.2, Figure 4.3).

Comparison	$\lvert\cos(\tilde{\mathbf{m}}_k,\, \mathbf{m}_k)\rvert$
$\tilde{\mathbf{m}}_1$ vs. $\mathbf{m}_1$	0.988
$\tilde{\mathbf{m}}_2$ vs. $\mathbf{m}_2$	0.980
$\tilde{\mathbf{m}}_3$ vs. $\mathbf{m}_3$	0.931

Table 4.2: Absolute cosine between the pooled-PCA components $\tilde{\mathbf{m}}_k$ and the sign-aligned-mean meta-axes $\mathbf{m}_k$ at layer 13. The subspace mutual-capture rate across the top-3 span is 96.6%.

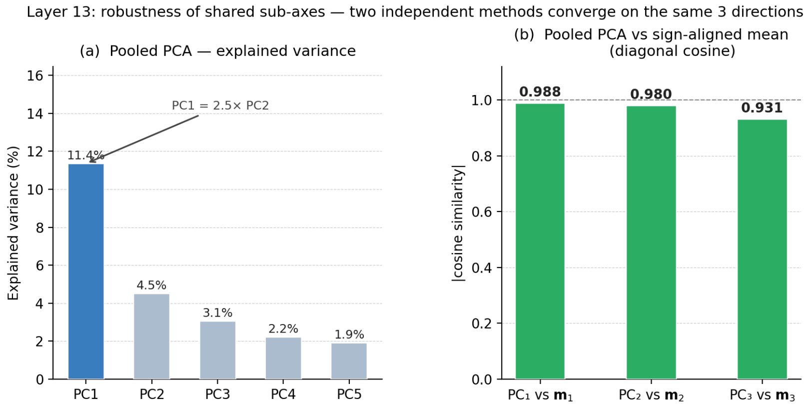

Figure 4.3: Convergence of pooled PCA and sign-aligned-mean meta-axes at layer 13. **(a)** Scree plot: PC_1 accounts for 11.4% of total variance, over $2.5\times$ the second component, indicating one strongly dominant shared direction. **(b)** Diagonal cosine between each pooled component $\tilde{\mathbf{m}}_k$ and the corresponding meta-axis $\mathbf{m}_k$; all three values exceed 0.93.

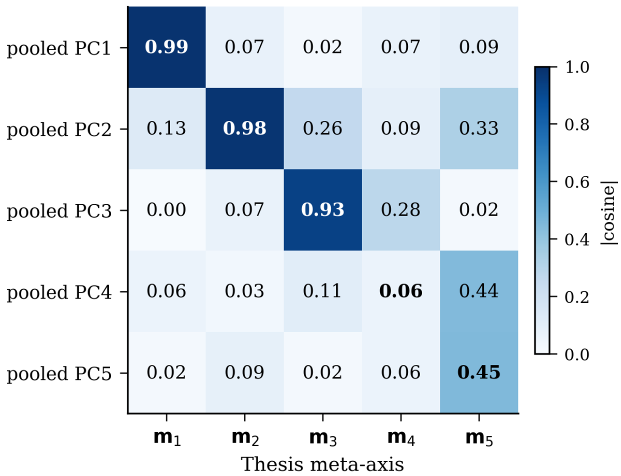

Figure 4.4: Full 5×5 absolute cosine matrix between pooled PCA components and the sign-aligned-mean meta-axes $\mathbf{m}_k$ at layer 13. The top-left 3×3 block is strongly diagonal (diagonal ≥ 0.93), confirming that the correspondence is axis-level, not merely subspace-level. Rows PC_4–PC_5 show no substantial alignment above 0.50 with any $\mathbf{m}_k$.

The off-diagonal entries of the full 3×3 cosine matrix are mostly small, confirming that pooled PCA reproduces not just the shared subspace but each individual axis. The mutual subspace-capture rate across the top-3 span is 96.6%.

Geometric Properties of $\mathbf{m}_k$

The three meta-axes satisfy two further geometric properties. First, all $\mathbf{m}_k$ are nearly orthogonal to the shared emotionalisation direction $\mathbf{g}$ ($|\cos(\mathbf{m}_k, \mathbf{g})| < 0.02$

for all k). Second, they are nearly orthogonal to every per-emotion prototype $\hat{\mathbf{r}}_e$ ($|\cos(\mathbf{m}_k, \hat{\mathbf{r}}_e)| < 0.09$ for all k, e). The second property is not incidental: it follows directly from the extraction procedure, in which step 3 projects out $\hat{\mathbf{r}}_e$ before PCA is applied, confining the $\mathbf{m}_k$ to the subspace orthogonal to the prototype by construction. Appendix A.9 (A1) provides supporting evidence that this step is load-bearing: without prototype removal the leading pooled component collapses onto the global emotionalisation direction. Each meta-axis therefore captures not emotion identity but affective texture: the continuous variation in how a discrete emotion is expressed, independent of which emotion is active. Together, these properties confirm that the $\mathbf{m}_k$ basis is geometrically independent of both the global emotionalisation axis and the emotion-label directions, recovering the within-emotion variance that prototype averaging discards.

Components $\tilde{\mathbf{m}}_4$ and $\tilde{\mathbf{m}}_5$ exhibit lower cross-method stability: each explains under 2.2% of variance, mutual-capture rates with the corresponding sign-aligned-mean axes fall below 0.50, and PC_4–PC_5 show cosines ≤ 0.45 against all $\mathbf{m}_k$ (Figure 4.4). We therefore distinguish the three stable axes ($\mathbf{m}_1$, $\mathbf{m}_2$, $\mathbf{m}_3$) from the two low-stability axes ($\mathbf{m}_4$, $\mathbf{m}_5$). Although $\mathbf{m}_4$ and $\mathbf{m}_5$ are geometrically low-stability, all five axes are carried forward for interpretation and causal testing in Sections 4.3–4.4: whether each axis is causally operative is an empirical question resolved by the steering experiment rather than by geometric stability alone.

4.2.4 Robustness: Convergence of Two Independent Methods

The convergence reported in Table 4.2 and Figure 4.3 is the outcome of a deliberate robustness test. This subsection makes the design of that test explicit and states what the convergence implies.

The sign-aligned mean and pooled PCA represent fundamentally different extraction strategies. The sign-aligned mean partitions the data by emotion label first: it runs a separate PCA within each of the eight emotion categories, then sign-aligns and averages the results rank by rank, relying on the emotion partition at every stage. Pooled PCA, by contrast, centres each emotion block to remove its mean and then applies a single decomposition to the stacked residuals. The centring step uses emotion labels, but the PCA itself operates on an unlabelled matrix and never consults which observation belongs to which emotion. The two methods therefore encode independent structural assumptions about how shared directions should be discovered.

One natural concern is that the meta-axes could be an artefact of the sign-aligned-mean procedure: for instance, that the sign-alignment step or the per-emotion partition imposes structure that would not otherwise be present. Pooled PCA is immune to both the sign-alignment choice and rank-wise recombination. After block-centring, it applies a single decomposition to the stacked matrix without consulting emotion identity during the eigendecomposition. Yet it recovers the same three directions, matching each $\mathbf{m}_k$ individually at the axis level and collectively at the subspace level (Table 4.2, Figures 4.3–4.4), ruling out those methodological choices as the source of $\mathbf{m}_1$, $\mathbf{m}_2$, and $\mathbf{m}_3$.

Two methods that share no internal steps converge on the same three affective axes. This convergence constitutes the strongest available evidence that $\mathbf{m}_1$, $\mathbf{m}_2$, and $\mathbf{m}_3$ reflect genuine shared structure in the model's residual activations rather than an artefact of any particular extraction procedure.

4.3 Pre-Verbal Interpretation of Meta-Axes

We now turn to the question of what affective information the shared meta-axes $\mathbf{m}_k$ encode. Unlike the residual vectors $\hat{\mathbf{r}}_e$, which are indexed by named emotion categories, the $\mathbf{m}_k$ are extracted from variance structure shared across emotions. Each $\mathbf{m}_k$ is not indexed to any single emotion label such as *joy*, *sadness*, or *anger*; instead, we treat it as a candidate **pre-verbal affective dimension**: a direction that modulates affective experience or expression prior to discrete category assignment. We interpret all five axes, including the low-stability axes $\mathbf{m}_4$ and $\mathbf{m}_5$. Labels for these two axes are provisional, and causal activity is assessed for each axis independently in Section 4.4.

To interpret these directions, we apply a contrastive LLM evaluator (`GPT-4o`) procedure. For each meta-axis, we collect examples with extreme positive and negative projections pooled across all eight emotions, then present all five axes to the judge simultaneously, instructing it to assign distinct labels to the high and low poles of each axis. Simultaneous presentation is critical: when axes are interpreted in isolation, the dominant signal tends to collapse onto a generic engaged-vs-detached or arousal-like distinction. By requiring the evaluator to compare axes against one another, the procedure elicits labels that capture what is specific to each axis rather than what is shared by all affective variation. The results are shown in Table 4.3.

A preliminary version presented each axis to the judge independently; this collapsed onto nearly identical labels for all axes (*engaged vs. detached* or *aroused vs. calm*), confirming that independent presentation is insufficient to elicit axis-specific labels. We note that the labels for all axes, and particularly for the low-stability axes $\mathbf{m}_4$ and $\mathbf{m}_5$, may partly reflect surface regularities in the generated text (topic, phrasing, or style) rather than latent affective structure. The contamination ratings reported by the judge should be consulted alongside the labels themselves.

For each axis, the judge returned an axis name, high- and low-pole descriptions, a short pre-verbal interpretation, a contamination rating, and a confidence score. The contamination rating estimates whether the contrast reflects affective content or may partly track topic, phrasing, or stylistic regularities in the generated text. The confidence score reflects how consistently the contrast was observed across pooled examples.

Axis	Axis label	Pre-verbal interpretation	Conf.
$\mathbf{m}_1$	Agitated/reactive **vs.** mellow/accepting	Arousal and affective reactivity: how strongly and immediately an affective state is felt.	5
$\mathbf{m}_2$	Nostalgic/ruminative **vs.** pragmatic/forward-looking	Temporal orientation: backward-looking rumination versus present-focused pragmatic coping.	4
$\mathbf{m}_3$	Interpersonally engaged / closure-seeking **vs.** self-contained / processing	Social versus internal orientation: whether affect is directed outward or processed internally.	4
$\mathbf{m}_4$	Mundane/obligatory **vs.** meaningful/fulfilling	Existential weight: trivial routine versus personally significant experience.	4
$\mathbf{m}_5$	Relief/release **vs.** tension/constraint	Somatic pressure: physical or situational release versus constriction.	5

Table 4.3: LLM-judge interpretation of the five meta-axes at layer 13. Labels are derived from contrastive evaluation across all eight named emotions simultaneously, yielding emotion-independent descriptions.

These axes suggest that the shared residual subspace is organised not only by emotion category but also by cross-cutting affective factors. $\mathbf{m}_1$ captures the most stable of these: an arousal-like distinction between agitated, reactive engagement and calmer acceptance. The remaining axes partition this affective space along more specific dimensions: temporal orientation ($\mathbf{m}_2$), interpersonal versus internal processing ($\mathbf{m}_3$), appraised significance ($\mathbf{m}_4$), and relief versus constraint ($\mathbf{m}_5$).

Each axis finds a counterpart in established theoretical frameworks. $\mathbf{m}_1$ corresponds to the arousal dimension in Russell's circumplex model [34]; $\mathbf{m}_2$ parallels the ruminative-versus-forward-oriented distinction in response-styles accounts of emotion regulation [35]; $\mathbf{m}_3$ parallels the social sharing and interpersonal regulation of emotion [36]; $\mathbf{m}_4$ parallels appraisal-theoretic notions of relevance and goal-conduciveness [37]; and $\mathbf{m}_5$ aligns with somatic and situational accounts of affective constraint [38].

These correspondences are interpretive rather than identificatory: the axes are not validated measurements of psychological constructs, and the LLM-judge labels may partly reflect surface regularities in the generated text rather than latent affective structure. The stable axes $\mathbf{m}_1$–$\mathbf{m}_3$ yield consistent cross-method labels; labels for $\mathbf{m}_4$ and $\mathbf{m}_5$, which are geometrically less stable, should be treated as provisional. We conclude not that the model implements a particular psychological theory, but that its affective residual space contains label-independent dimensions whose poles resemble established pre-verbal factors of emotional experience. Whether these dimensions are causally operative, that is, whether injecting them into the residual stream produces the predicted expressive shifts, is addressed in Section 4.4.

Named emotions in this representation are not atomic: a generated expression of *anger*, *fear*, or *joy* can vary independently along multiple affective axes. This supports a compositional view in which the emotion label specifies the prototype target while the meta-axes modulate how that prototype is expressed.

4.4 Causal Validation: β Sweep Along Meta-Axes

The geometric analysis in the preceding sections established two structural properties. First, the $\mathbf{m}_k$ are orthogonal to both the global emotionalisation direction g and the per-emotion prototypes $\hat{\mathbf{r}}_e$. Second, their LLM-assigned poles resemble established pre-verbal affective dimensions.

However, structural geometry alone does not establish causal relevance. To make the stronger claim that the $\mathbf{m}_k$ are causally active representations, rather than correlated geometric artefacts, we need a direct intervention. Specifically, we inject $\beta, \mathbf{m}_k$ into the model's residual stream. We then test whether the expressed affective texture shifts in the predicted direction. Finally, we verify that this shift occurs at the texture level rather than at the emotion-category level.

4.4.1 Protocol

Following the activation-steering protocol of Section 3.8.1, we apply forward-hook steering at layer 13 across all five meta-axes, injecting a single steering vector

$$\Delta = \beta\, \mathbf{m}_k$$

at the final token position with no emotion-prototype component ($\alpha_R = 0$). The sign convention is fixed per axis: positive β steers toward the first-named pole in the axis label (e.g. *agitated/reactive* for $\mathbf{m}_1$), and negative β steers toward the second-named pole. The scale parameter is swept over $\beta \in \{-16, -12, -8, -4, 0, 4, 8, 12, 16\}$, with $\beta = 0$ serving as the unsteered baseline. Generation uses greedy decoding with `max_new_tokens=60`. We evaluate the first 100 source texts from the rewrite-database test split, yielding $100 \times 9 \times 5 = 4{,}500$ generated outputs.

Each output is scored by a `GPT-4o-mini` judge on four metrics (all $\in [0, 1]$):

- **soundness**: fluency, grammaticality, and absence of repetition;
- **meaning_preserved**: retention of the propositional content of the source;
- **axis_pole_match** (hereafter `pm`): shift toward the expected pole of $\mathbf{m}_k$ given the sign of β;
- **subtle_affective_shift**: change in expressive mode without change in emotion label.

The judge is provided with the axis name, both pole descriptions, and the expected steering direction, but not with the value of β or the source text. Because the judge scores **subtle_affective_shift** as a change in expressive mode *without* a change in emotion label, high scores on this metric alongside high `pm` serve as joint evidence that the observed shift is texture-level rather than category-level. Coherence window bounds L_β and R_β are defined as the outermost β values at which `soundness` ≥ 0.70 still holds.

4.4.2 Results

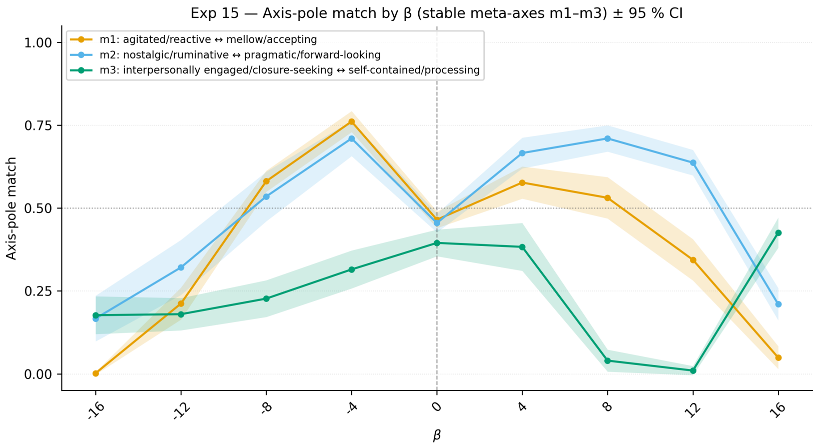

Figure 4.5: Axis-pole match rate as a function of steering coefficient β for the three stable meta-axes $\mathbf{m}_1$–$\mathbf{m}_3$ ($N = 100$, 95% CI). The vertical dashed line indicates the unsteered baseline ($\beta = 0$). While $\mathbf{m}_1$ and $\mathbf{m}_2$ peak sharply at $\beta = -4$, $\mathbf{m}_3$ remains near the unsteered baseline throughout the coherent steering window.

Figure 4.5 shows that both $\mathbf{m}_1$ and $\mathbf{m}_2$ exhibit a robust steering response. Here, $\mathbf{m}_1$ corresponds to agitated/reactive $\leftrightarrow$ mellow/accepting. $\mathbf{m}_2$ corresponds to nostalgic/ruminative $\leftrightarrow$ pragmatic/forward-looking.

Both axes sit near chance at the unsteered baseline. The baseline pole-match scores are pm@0 $= 0.46$–0.47. Steering at $\beta = -4$ raises pole match to 0.761 for $\mathbf{m}_1$ and 0.710 for $\mathbf{m}_2$. These correspond to absolute gains of $+0.26$–$+0.30$.

The effect occurs within coherent steering windows. These windows are $[-8, 16]$ for $\mathbf{m}_1$ and $[-16, 12]$ for $\mathbf{m}_2$. Across both windows, soundness remains ≥ 0.94. Meaning preservation also remains ≥ 0.97. This indicates that the affective shift does not degrade fluency or propositional content.

The `subtle_affective_shift` score closely tracks pole match throughout the sweep. This confirms that the observed change reflects a pre-verbal expressive shift. It is therefore better understood as a change in affective texture, rather than as an emotion-category substitution.

We find only a marginal gain for $\mathbf{m}_3$ (interpersonally engaged $\leftrightarrow$ self-contained/processing): $\mathtt{pm}_{\text{peak}} = 0.426$ at $\beta = +16$, just $+0.031$ above the unsteered baseline of 0.395 (Figure 4.5). We speculate that interpersonal orientation is not reliably accessible to single-text continuation steering at layer 13, consistent with the near-flat pole-match curve across the entire coherent window $\beta \in [-12, 16]$.

Axis	**Stability**	L_β	R_β	M_A	pm@0	$\mathtt{pm}_{\text{peak}}$	β_{peak}
$\mathbf{m}_1$	stable	-8	16	4.0	0.465	**0.761**	-4
$\mathbf{m}_2$	stable	-16	12	-2.0	0.455	**0.710**	-4
$\mathbf{m}_3$	stable	-12	16	2.0	0.395	0.426	$+16$
$\mathbf{m}_4$	low	-16	8	-4.0	0.345	0.563	-4
$\mathbf{m}_5$	low	-8	12	2.0	0.420	**0.812**	$+4$

Table 4.4: Coherent window (L_β, R_β), window centre M_A, baseline pole match pm@0, peak pole match $\mathtt{pm}_{\text{peak}}$, and optimal β for each meta-axis ($N = 100$). Bold entries indicate axes with $\mathtt{pm}_{\text{peak}} \geq 0.70$.

Despite its low-stability flag at extraction, $\mathbf{m}_5$ (relief/release $\leftrightarrow$ tension/constraint) achieves the highest $\mathtt{pm}_{\text{peak}}$ of any axis at 0.812 ($\beta = +4$; Table 4.4). We speculate that the $N = 100$ texts, which contain naturalistic descriptions of situational and bodily states, align particularly well with this somatic dimension, yielding an unusually clean steering signal. The coherent window is $\beta \in [-8, 12]$.

$\mathbf{m}_4$ (mundane/obligatory $\leftrightarrow$ meaningful/fulfilling) reaches $\mathtt{pm}_{\text{peak}} = 0.563$ at $\beta = -4$ (LOW pole; Table 4.4), while the HIGH pole ($\beta > 0$) yields near-zero pole match, revealing a marked asymmetry between poles. This suggests that the label assignment for $\mathbf{m}_4$ may be sign-inconsistent with its causal effect, which is unsurprising given its low cross-method stability.

Across all axes, the coherent window is confined to $|\beta| \lesssim 12$–16 (Table 4.4). At the boundaries, soundness degrades sharply (soundness $= 0.046$ for $\mathbf{m}_1$ at $\beta = -16$; soundness $= 0.041$ for $\mathbf{m}_4$ at $\beta = 16$). This degradation provides an empirical upper bound on the usable steering magnitude and motivates the use of $|\beta| \leq 8$ as a conservative default.

4.4.3 Qualitative Results

Table 4.5 presents representative examples of how a seed sentence expressing *fear* is modulated by steering along $\mathbf{m}_1$, the *agitated/reactive* $\leftrightarrow$ *mellow/accepting* axis. Three patterns are visible.

Positive β increases the visceral intensity of *fear* without altering the emotion label. At $\beta = 4$, the propositional content is nearly identical to the unsteered baseline ($\beta = 0$). The same creaking floorboard remains. The same old mansion remains. The same inability to shake the unease also remains. What changes is the affective register. The phrase *on edge* and the construction *utterly unable to shake off* make the response more immediate and bodily than the baseline's *breathless and unable to shake off the feeling*. At $\beta = 8$, the sensation escalates further. The floorboards *reverberate through every fiber of my being*. *Trepidation courses through my very soul*. Anxiety also *refuses to subside*. *fear* is no longer merely a situational alert. It becomes an unrelenting somatic state. This pattern is categorically distinct from the $\hat{\mathbf{r}}_e$ steering analysed in Section 3.8.5. Increasing α_R shifts the emotion category, for example from *sadness* to *anger*. By contrast, increasing β shifts affective texture within a fixed emotion. The shift is from contained unease towards visceral reactive terror.

Condition	**Steered output**
Base: Every creak of the floorboard made my heart race and I could not calm down.	
$\beta = -4$	The eerie silence was shattered by every creak of the floorboard, making my heart skip a beat and my anxiety spike, but I couldn't shake the feeling that I was being watched.
$\beta = 0$	As I stepped into the old mansion, every faint creak of the floorboard beneath my feet sent a jolt of anxiety coursing through my veins, leaving me breathless and unable to shake off the feeling of unease that had taken hold of me.
$\beta = 4$	As I stepped into the old mansion, the faintest creak of the floorboards beneath my feet sent a jolt of anxiety coursing through my veins, leaving me on edge and utterly unable to shake off the sense of unease that had taken hold of me.
$\beta = 8$	The mere sound of the floorboards beneath my feet seemed to reverberate through every fiber of my being, sending a jolting sense of trepidation coursing through my very soul, leaving me utterly consumed by an unrelenting sense of anxiety that refused to subside.
$\beta = 16$	It seems that the statement is conveying that you are experiencing intense emotions, but it is clear that the statement is expressing your feelings in a concise and straightforward manner.

Table 4.5: A base sentence "Every creak of the floorboard made my heart race and I could not calm down" steered with $\mathbf{m}_1$ (*agitated/reactive* $\leftrightarrow$ *mellow/accepting*). Positive β steers toward the agitated/reactive high pole; negative β steers toward the mellow/accepting low pole.

Negative β steers the output towards the mellow/accepting pole while preserving the emotion label. At $\beta = -4$, the same situation obtains: the floorboard creaks, the speaker cannot shake the feeling of being watched. Yet the reaction is qualitatively softened. The original sentence's *"heart race"* becomes *"heart skip a beat"*; the anxiety is acknowledged as a *spike* but is immediately followed by the resigned concessive *"but I couldn't shake the feeling that I was being watched"*. The *fear* is present but accepted rather than resisted, a distinction that does not correspond to any change in the named emotion. This supports the interpretation that $\mathbf{m}_1$ functions as a pre-verbal texture axis: it modulates how an emotion is held and expressed, not which emotion is present. The contrast with $\hat{\mathbf{r}}_e$ steering is instructive: negative α_R suppresses the target emotion and redirects affect towards an opposing category, whereas negative β shifts the same *fear* towards a more resigned, accepting quality without category substitution.

Excessive intervention degrades generation quality through a failure mode distinct from that observed with $\hat{\mathbf{r}}_e$. At $\beta = 16$, the output produces meta-commentary rather than a steered sentence: *"It seems that the statement is conveying that you are experiencing intense emotions"*. The model exits first-person experiential mode entirely and narrates from outside the representational regime. This collapse of perspective is qualitatively different from the lexical repetition loops observed at high α_R. Quality visibly degrades by $\beta = 16$, consistent with the broader sweep showing that extreme values near the edge of the coherent window reduce soundness and meaning preservation.

4.4.4 Interpretation

Three conclusions follow from the steering experiment. First, the meta-axes are causally active. Directly injecting $\beta, \mathbf{m}_k$ produces consistent, judge-verified shifts in affective texture at moderate steering magnitudes.

Second, the effect is texture-specific. The `pm` score closely tracks `subtle_affective_shift`. At the same time, meaning preservation remains high inside the coherent window. This indicates that the shift is in the how of expression rather than the what.

Third, the causal effect is axis-dependent. Although $\mathbf{m}_3$ is geometrically stable, it resists steering throughout the coherent window. This suggests that geometric recoverability does not guarantee single-layer causal controllability. We speculate that interpersonal orientation is less localised at a single residual-stream layer than arousal or temporal orientation. Confirming this would require multi-layer ablations beyond the scope of the current experiments. Appendix A.9 (A2) discusses the extent to which the $\beta = 0$ baseline serves as a partial falsification control and identifies a random-direction injection experiment as the outstanding validation step.

These results complete the argument of this chapter. At least $\mathbf{m}_1$ and $\mathbf{m}_2$ are not merely geometric structures correlated with pre-verbal affective dimensions. In the present dataset, the low-stability axis $\mathbf{m}_5$ also behaves causally. These axes can be independently manipulated to shift the texture of emotional expression without altering the emotion label.

The steering formula is therefore

$$\Delta_e \;=\; \alpha_R\,\hat{\mathbf{r}}_e \;+\; \sum_k \gamma_k\,\mathbf{m}_k,$$

where the two terms are functionally separable. The coefficient α_R controls emotion identity. The coefficients γ_k modulate pre-verbal affective texture.

Chapter 5

Evaluation

The two preceding chapters each presented a self-contained experiment with its own quantitative and qualitative assessments. This chapter steps back from both and evaluates the project as a whole: what the results collectively establish, where the evidence falls short, and what limitations remain open.

5.1 Summary of Results Across Experiments

The two experiments (Chapters 3 and 4) address complementary questions about the structure and controllability of affective representations in an LLM.

Chapter 3 showed that raw CAA vectors are dominated by a shared emotionalisation component $\mathbf{g}$, and that projecting it out yields emotion-specific residuals $\hat{\mathbf{r}}_e$ that are more discriminative, better recover Plutchik's circular geometry, and produce reliable affective shifts during inference. The shared component is analytically useful for isolating emotion-specific signal but counterproductive as a generation-time intervention.

Chapter 4 showed that the within-emotion residual subspace is not flat: named emotions vary substantially in internal dimensionality, and pooled PCA recovers three stable shared axes confirmed by two independent extraction methods. Two of these axes ($\mathbf{m}_1$, $\mathbf{m}_2$) are causally active, producing judge-verified affective texture shifts without altering the emotion label, while meaning preservation remains high throughout the coherent steering window.

5.2 Human Evaluation Study

To validate that the affective shifts detected by the automated LLM judge are perceptible to human readers, we conduct a small-scale human evaluation study ($N = 18$ participants) using the form described in Appendix A.7. The study covers three tasks corresponding to the three central claims of the thesis.

5.2.1 Part 1: Perceptibility of Meta-Axis Texture Shifts

Part 1 of the study tests whether the $\mathbf{m}_1$-steered texture shifts are perceptible to human readers. What we want to focus on here is, while the emotion category remains the same (e.g. joy remains joy, sadness remains sadness), whether one sentence is perceived by humans as

- more physically overwhelming
- expressing emotion in a more direct and instinctive way
- conveying a sense of urgency or raw intensity

than the other.

Participants are presented with pairs of sentences generated from the same source utterance, one steered toward the *agitated/reactive* pole ($\beta = +4$) and one toward the *mellow/accepting* pole ($\beta = -4$), without being told which is which. For each pair, participants select which sentence feels more intense and immediate in its emotional expression. This is a two-alternative forced choice (2AFC) task, meaning that participants must choose one of the two sentences as more intense, with no neutral option. Formally, the hypothesis is that if the $\mathbf{m}_1$ direction captures an *agitated / immediate* sub-axis within an emotion, then human evaluators should consistently select the $\beta = +4$ output as more emotionally intense and immediate than the comparison output.

Q	**Emotion**	**Expected Direction**	**Correct ($N = 18$)**	**Proportion**	**p-value (binomial)**
Q1	Joy	Sentence 2	14	0.778	0.031^{*}
Q2	Fear	Sentence 1	14	0.778	0.031^{*}
Q3	Disgust	Sentence 2	14	0.778	0.031^{*}
Q4	Anger	Sentence 1	17	0.944	$< 0.001^{***}$
Q5	Sadness	Sentence 2	17	0.944	$< 0.001^{***}$
Overall	—	—	76	0.844	$< 0.001^{***}$

$^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$ (exact two-sided binomial test against chance $p_0 = 0.5$).

Table 5.1: Part 1: Pairwise intensity discrimination results ($N = 18$). Participants selected which of two sentences expressed more emotional intensity. Expected direction indicates the $\beta = +4$ (agitated pole) steered sentence.

Across all 90 judgements ($N = 18$, 5 pairs each), the expected sentence was selected on 76 occasions (84.4%), well above the 50% chance level. The pooled exact binomial test confirmed this effect ($p < 0.001$). This provides strong evidence that the $\mathbf{m}_1$-steered texture shifts are perceptible to human readers as intended. The effect was particularly consistent for *anger* and *sadness*, where 17 of 18 participants selected the expected sentence ($p < 0.001$ each). Joy, fear, and disgust showed somewhat lower but still significant agreement ($14/18$, $p = 0.031$ each).

5.2.2 Part 2: Emotion Category Preservation

Part 2 verifies that $\mathbf{m}_1$-steered texture shifts do not alter the perceived emotion category. While Part 1 asked participants to select the more intensity-rich sentence, Part 2 asks them to assign a specific emotion label to each sentence. Formally, we verify that varying β does not substitute one emotion category for another, but shifts representational style within the same category. Specifically, if sentences at both $\beta = -4$ and $\beta = 0$ for *fear* are judged to express *fear*, this implies that the β operation modulates expressive texture rather than converting the emotion itself.

Q	**Target Emotion**	**Condition (β)**	**Correct ($N = 18$)**	**Proportion (%)**
Q6	Fear	$\beta = -4$	17	94.4
Q7	Fear	$\beta = 0$	17	94.4
Q8	Anger	$\beta = +4$	16	88.9
Q9	Anger	$\beta = -4$	15	83.3
Q10	Sadness	$\beta = +4$	17	94.4
Q11	Sadness	$\beta = -4$	17	94.4
Overall	—	—	99	91.7

Table 5.2: Part 2: Emotion identification results ($N = 18$). Participants labelled each steered sentence with one of Plutchik's eight basic emotions. β denotes the axis-steering magnitude applied to the $\mathbf{m}_1$ direction.

Category agreement with the intended emotion was high across all conditions (Table 5.2): 94.4% for *fear* at both $\beta = -4$ and $\beta = 0$, and 94.4% for *sadness* at both $\beta = +4$ and $\beta = -4$. Agreement for *anger* was 88.9% at $\beta = +4$, with one participant selecting *disgust* and one selecting *none of the above*, and 83.3% at $\beta = -4$, with two participants selecting *sadness* and one selecting *trust*.

Category labels remained stable across β values within each emotion. Overall, 99 of 108 responses (91.7%) matched the intended emotion. These results confirm that $\mathbf{m}_1$ steering shifts expressive texture without substituting the emotion category, consistent with the `subtle_affective_shift` scores reported by the LLM judge in Chapter 4.

5.2.3 Part 3: Perceptibility of Prototype Steering

To measure whether stronger target-emotion steering produces a perceptible increase in target-emotion intensity, Part 3 presents participants with pairs of sentences and asks them to rate each sentence independently. While Parts 1 and 2 use forced-choice comparisons, Part 3 uses an independent Likert rating for each sentence, decoupling the two ratings so that participants assign an absolute intensity score on its own merits rather than selecting the more intense of a pair.

Specifically, each evaluation set consists of an original utterance, a designated target emotion, an unsteered sentence ($\alpha_R = 0$), and a sentence steered at $\alpha_R = 64$, presented without any indication of which α_R value produced them. For each sentence, participants rate *How strongly do you feel [target emotion] in this sentence?* on a five-point scale (1 = *do not feel it at all*; 5 = *feel it very strongly*). The scale captures perceived target-emotion intensity and does not ask participants to assess naturalness or overall quality.

The central hypothesis is that sentences generated with stronger target-emotion steering should receive higher human-rated intensity than the unsteered baseline, i.e., $\mu_{\alpha_R=64} > \mu_{\alpha_R=0}$ for each of the six sentence pairs.

Set	**Target emotion**	$\mu_{\alpha=64}$ **(SD)**	$\mu_{\alpha=0}$ **(SD)**	Δ	$t(17)$
Anger set 1	Anger	3.56 (0.86)	1.89 (0.90)	$+1.67$	6.52^{***}
Anger set 2	Anger	3.61 (1.09)	1.67 (0.69)	$+1.94$	8.26^{***}
Fear set 1	Fear	3.72 (0.96)	1.78 (0.81)	$+1.94$	6.81^{***}
Fear set 2	Fear	3.89 (1.18)	1.78 (0.73)	$+2.11$	8.30^{***}
Joy set 1	Joy	4.17 (0.51)	2.33 (0.97)	$+1.83$	7.08^{***}
Sadness set 1	Sadness	4.39 (0.50)	2.06 (0.42)	$+2.33$	14.43^{***}
Overall		**3.89**	**1.92**	$+$**1.97**	—

$^{***}p < 0.001$ (two-sided paired t-test, $df = 17$). Ratings are on a 1–5 Likert scale.

Table 5.3: Part 3: Mean human intensity ratings for steered ($\alpha_R = 64$) and unsteered ($\alpha_R = 0$) sentences across six emotion–source pairs ($N = 18$). All differences are in the expected direction.

We find that the steered sentence receives a higher mean rating than the baseline in every pair (Table 5.3), with an overall mean difference of $\Delta = +1.97$ points. Paired t-tests confirm the effect is highly significant for each individual pair ($t(17) \geq 6.52$, $p < 0.001$ in all cases).

The Spearman correlation between LLM-judge `target_emotion_intensity` scores and human mean ratings, computed across 12 conditions (six pairs $\times$ two α_R levels), is $\rho = 0.643$, indicating moderate positive agreement. The main discrepancy arises in Fear set 2 at $\alpha_R = 64$: the LLM judge assigns a score of 0.00 while humans rate the same sentence at 3.89, suggesting the automated evaluator underestimates fear intensity in certain surface constructions. These results broadly support LLM-as-judge evaluation as a proxy for human perception, though the correspondence is imperfect and human calibration remains important.

Together, these three tasks show that the proposed steering directions correspond to affective differences that human readers can reliably perceive, corroborating the automated evaluation results and partially addressing the LLM-as-a-judge limitation identified in Section 5.3.

5.3 Limitations

5.3.1 LLM-as-a-Judge Evaluation

Both experiments use `GPT-4o` or `GPT-4o-mini` as the primary evaluator [39]. One limitation of this setup is that the judge and the steered model (`Llama-3.1-8B-Instruct`) are both instruction-tuned on overlapping corpora, so the judge may encode the same surface-level stylistic regularities it is asked to assess, rather than evaluating independent affective signal.

A second concern is label coverage: the judge rates outputs against Plutchik's eight emotion labels and the axis pole descriptions supplied in the prompt; affective shifts that fall outside this label space may be scored incorrectly or collapsed onto the nearest label.

A third concern is external validity. Judge scores are continuous proxies for human perception, and the human validation study ($N = 18$) is too small to establish ground-truth calibration. These findings should therefore be interpreted as LLM-perceived affective shifts. Whether they correspond to shifts perceived by human readers remains to be established with a larger-scale study.

5.3.2 Single-Layer Intervention

All steering experiments inject vectors at a single transformer layer (layer 13), following the layer diagnostics in Section 3.4. This is a principled but simplifying choice. Affective processing need not be localised at a single layer, and different dimensions may be best steered at different depths. The near-zero steering effect observed for $\mathbf{m}_3$ (interpersonal engagement) is consistent with this possibility: the axis may be distributed across layers in a way that a single-site injection cannot perturb effectively. Multi-layer interventions and per-axis layer selection were not explored in this work. A systematic study varying injection depth per axis would be needed to determine whether the null effect for $\mathbf{m}_3$ reflects representational geometry or simply a mismatch between the intervention site and where that axis is encoded.

5.3.3 Scope of Emotions and Generalisability

The CAA vectors are defined by contrastive pairs drawn from Plutchik's eight basic emotions, a choice that constrains the geometry of the resulting representation space. This taxonomy is a theoretical commitment: other frameworks (e.g. Russell's valence-arousal circumplex, or GEMS for music-evoked states) would partition emotion space differently, producing different contrasting pairs and, consequently, different CAA directions. Whether the decomposition $\mathbf{g}$, $\hat{\mathbf{r}}_e$, $\mathbf{m}_k$ is specific to Plutchik's categories or reflects a more general property of affective geometry is an open question.

A further confound is introduced by the data generation procedure. The rewrite dataset is constructed from English dialogue corpora using GPT-4.1-mini. Affective shifts captured in the residual stream may therefore partly reflect GPT-4.1-mini's stylistic regularities rather than the inherent affective geometry of `Llama-3.1-8B`. Disentangling the contribution of the generation model from the target model's internal representations would require a controlled study varying the rewrite model while holding the target model fixed.

All experiments use a single model family (`Llama-3.1-8B-Instruct`). Whether the decomposition generalises to other architectures or model scales is untested.

5.3.4 Interpretation of Meta-Axis Labels

The labels assigned to $\mathbf{m}_1$–$\mathbf{m}_5$ (Table 4.3) are post-hoc interpretations produced by prompting an LLM judge to characterise the contrast between pole-matched examples. Such labels may partly track surface regularities in the generated text (topic, register, phrasing) rather than latent affective structure, and should be treated as plausible characterisations rather than ground-truth annotations.

5.4 Critical Appraisal

The experiments provide substantial evidence for the proposed decomposition of affective steering directions, though the strength of that evidence varies across claims. This section distinguishes findings directly supported by the experiments from interpretations that warrant more caution.

The strongest empirical claim is that emotion-related CAA directions contain a large shared component and a smaller emotion-specific residual. Several converging results support this. The raw CAA vectors are highly similar to one another, suggesting that much of the signal reflects a general shift from neutral to emotional language rather than encoding any single named emotion. After removing the shared direction g, the residual vectors $\hat{\mathbf{r}}_e$ become more useful for distinguishing emotion categories and better reflect the expected relationships among Plutchik emotions. These residual directions are also causally active: injecting them during generation produces predictable target-emotion shifts. $\hat{\mathbf{r}}_e$ is therefore not merely a descriptive artefact but a functional steering direction.

A second well-supported claim is that affective steering is not exhausted by choosing a discrete emotion label. Chapter 4 shows that there is meaningful structure within each named emotion: the meta-axes $\mathbf{m}_1$ and $\mathbf{m}_2$ modulate the expressive texture of an emotion while largely preserving the emotion category. The human evaluation strengthens this point, as participants perceived the $\mathbf{m}_1$ shift as a difference in intensity and immediacy while still assigning the same intended emotion label in most cases. This suggests that the model encodes not only *which* emotion is expressed but also *how* it is expressed. The term *pre-verbal* is used in a narrow technical sense: $\mathbf{m}_k$ is sub-label, not pre-linguistic — the model has no phenomenal experience, and what is recovered is activation geometry that correlates with psychologically grounded affective dimensions.

The use of LLM-as-judge evaluation requires caution. The automated judge enabled consistent evaluation across many generations and multiple steering strengths, and the human evaluation broadly supports the automated results, especially for the perceptibility of $\mathbf{m}_1$ and the increase in target-emotion intensity under prototype steering. However, the correlation between LLM-judge scores and human ratings is only moderate, and a clear discrepancy arises in the fear condition. The LLM judge is therefore a useful scalable proxy but not a complete substitute for human evaluation. The strongest claims in this thesis are accordingly those supported by both sources of evidence.

The experimental design is deliberately controlled: greedy decoding, held-out source splits, fixed layers, and structured evaluation prompts reduce noise and isolate the causal effects of the vectors. This control is both a strength and a limitation. The experiments evaluate single-sentence rewriting from dialogue-like source utterances, and whether the same decomposition transfers to longer passages, multi-turn conversations, cross-lingual settings, or other genres remains to be established. The results should therefore be understood as evidence for a reliable mechanism within a constrained setting, not as proof of universal affective control.

Overall, the experiments support a bounded claim: affective generation in the studied model can be decomposed into a shared emotionalisation direction, emotion-specific residual directions, and sub-label texture axes that modulate expressive character within an emotion category. These components are geometrically meaningful, causally active, and perceptible to human readers. Whether the same decomposition is stable across model families, scales, or training regimes is an open question that the single-model, single-layer design cannot address.

Chapter 6

Conclusions and Future Work

6.1 Summary of Contributions

This project set out to decompose the affective residual stream of an LLM into interpretable, independently controllable components. The two experiments together establish a two-level structure:

$$\Delta_e = \alpha_R\,\hat{\mathbf{r}}_e + \sum_k \gamma_k\,\mathbf{m}_k,$$

where $\hat{\mathbf{r}}_e$ selects the emotion prototype and $\mathbf{m}_k$ modulates expressive texture independently of emotion identity. This decomposition is the primary contribution of the project.

The first experiment showed that raw CAA vectors conflate a shared emotionalisation component g with emotion-specific signal, and that projecting out g improves both classification accuracy and recovery of Plutchik's circular geometry. The second experiment showed that the within-emotion residual subspace is not flat: pooled PCA recovers stable shared axes that are causally active in generation yet leave the emotion category intact. Human evaluation corroborated both findings at perceptual scale.

Our main contributions are:

1. a geometric decomposition of CAA vectors that separates a shared emotionalisation component from emotion-specific prototypes, improving classification and recovering Plutchik's circular structure;

2. a pooled PCA method for extracting stable meta-axes from within-emotion residuals; $\mathbf{m}_1$ and $\mathbf{m}_2$ are causally active in generation without altering emotion category, while $\mathbf{m}_3$ is geometrically stable but not reliably steerable at layer 13; and

3. a human evaluation confirming that both decomposition levels produce perceptually distinct affective texture shifts.

6.2 Conclusions

The central question motivating this work was whether affective texture in LLM outputs can be decomposed and controlled independently of emotion category. The answer, at least within the scope tested here, is yes. The decomposition is geometrically principled and empirically validated. Steering along $\hat{\mathbf{r}}_e$ shifts emotion intensity, while steering along $\mathbf{m}_k$ shifts expressive texture without substituting the emotion label.

The meta-axes $\mathbf{m}_k$ are *pre-verbal* in a narrow technical sense: they are orthogonal to named emotion prototypes and recoverable from two independent extraction methods. Whether they correspond to dimensions of pre-linguistic affective experience is an interpretive claim that the current evidence supports but does not establish definitively. The model has no phenomenal experience; what is recovered is activation geometry that correlates with psychologically grounded affective dimensions.

Taken together, these results suggest that LLMs do not simply encode affect as a set of discrete labelled states. There is internal structure within each emotion that can be identified, measured, and manipulated, a finding with direct implications for affective computing applications and for mechanistic interpretability of emotional content in neural language models.

6.3 Future Work

Several directions follow directly from the limitations identified above.

The most immediate improvement would be to supplement LLM-as-judge scores with a larger human rater study measuring inter-rater agreement on affective texture shifts produced by meta-axis steering. A study of this kind would establish whether the judge-rated shifts are perceptually real at a scale beyond the present $N = 18$ pilot.

Additionally, extending the intervention to multiple layers simultaneously, or transferring the extracted vectors to different model scales, would test the robustness and generalisability of the decomposition. The failure of $\mathbf{m}_3$ to respond to single-layer steering motivates a systematic multi-layer ablation as a first step.

Jackson et al. [1] document substantial cross-cultural variation in emotion semantics. Testing whether the same affective geometry is recovered from multilingual corpora, and whether meta-axis steering produces consistent effects across languages, would directly address one of the core motivations stated in the Introduction 1.

Beyond Plutchik's taxonomy, the same methods could be applied to alternative emotion models. Repeating the construction with a different emotion taxonomy (e.g. Russell's circumplex VAD dimensions, or a data-driven label set) would test whether the meta-axes $\mathbf{m}_k$ reflect stable affective structure in the model or are artefacts of the Plutchik taxonomy.

Bibliography

1. Jackson JC, Watts J, Henry TR, List JM, Forkel R, Mucha PJ, Greenhill SJ, Gray RD, and Lindquist KA. Emotion semantics show both cultural variation and universal structure. Science 2019; 366:1517–22. DOI: 10.1126/science.aaw8160. Available from: https://www.science.org/doi/10.1126/science.aaw8160
2. Panickssery N, Gabrieli N, Schulz J, Tong M, Hubinger E, and Turner AM. Steering Llama 2 via Contrastive Activation Addition. 2024 Jul. Available from: http://arxiv.org/abs/2312.06681 [Accessed on: 2026 Jan 17]
3. Ekman P. Basic emotions. *Handbook of cognition and emotion*. Wiley, 1999 :45–60
4. Universal Emotions — What are Emotions? https://www.paulekman.com/universal-emotions/. Accessed: 2026-01-20
5. Mohammad SM, Bravo-Marquez F, Salameh M, and Kiritchenko S. SemEval-2018 Task 1: Affect in Tweets. *Proceedings of the 12th International Workshop on Semantic Evaluation*. 2018 :1–17
6. Demszky D, Movshovitz-Attias D, Ko J, Cowen A, Nemade G, and Ravi S. GoEmotions: A Dataset of Fine-Grained Emotions. Accessed: 2026-01-20. 2020. arXiv: 2005.00547 [cs.CL]. Available from: https://arxiv.org/abs/2005.00547
7. Wang X and Zong C. Distributed Representations of Emotion Categories in Emotion Space. *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*. Ed. by Zong C, Xia F, Li W, and Navigli R. Online: Association for Computational Linguistics, 2021 Aug :2364–75. DOI: 10.18653/v1/2021.acl-long.184. Available from: https://aclanthology.org/2021.acl-long.184/
8. Mehrabian A and Russell JA. An Approach to Environmental Psychology. Cambridge, MA: MIT Press, 1974
9. Mohammad SM. NRC VAD Lexicon v2: Norms for Valence, Arousal, and Dominance for over 55k English Terms. 2025. arXiv: 2503.23547 [cs.CL]. Available from: https://arxiv.org/abs/2503.23547 [Accessed on: 2026 Jan 12]

10. PAD emotional state model. Available from: https://en.wikipedia.org/wiki/PAD_emotional_state_model#/media/File:PAD_emotional_state_model_Page_1.png [Accessed on: 2026 Jan 20]

11. Buechel S, Modersohn L, and Hahn U. Towards Label-Agnostic Emotion Embeddings. 2021. arXiv: 2012.00190 [cs.CL]. Available from: https://arxiv.org/abs/2012.00190 [Accessed on: 2026 Jan 18]

12. Radford A, Jozefowicz R, and Sutskever I. Learning to Generate Reviews and Discovering Sentiment. 2017. arXiv: 1704.01444 [cs.LG]. Available from: https://arxiv.org/abs/1704.01444 [Accessed on: 2026 Jan 14]

13. Zhengxuan Wu YJ. Disentangling Latent Emotions of Word Embeddings on Complex Emotional Narratives. NLPCC2019 2019 Aug. Available from: https://arxiv.org/abs/1908.07817 [Accessed on: 2026 Jan 16]

14. Pennington J, Socher R, and Manning CD. GloVe: Global Vectors for Word Representation. *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics, 2014 :1532–43. DOI: 10.3115/v1/D14-1162. Available from: https://aclanthology.org/D14-1162

15. Plutchik R. Emotion: A Psychoevolutionary Synthesis. *Emotion: Theory, Research, and Experience*. Ed. by Plutchik R and Kellerman HR. New York, NY, USA: Academic Press, 1980

16. Pfau J, Merrill W, and Bowman SR. Let's Think Dot by Dot: Hidden computation in transformer language models. *First Conference on Language Modeling*. 2024. Available from: https://openreview.net/forum?id=NikbrdtYvG

17. Hao S, Sukhbaatar S, Su D, Li X, Hu Z, Weston J, and Tian Y. Training Large Language Models to Reason in a Continuous Latent Space. 2025. arXiv: 2412.06769 [cs.CL]. Available from: https://arxiv.org/abs/2412.06769 [Accessed on: 2026 Mar 14]

18. Shen Z, Yan H, Zhang L, Hu Z, Du Y, and He Y. CODI: Compressing Chain-of-Thought into Continuous Space via Self-Distillation. 2025. arXiv: 2502.21074 [cs.CL]. Available from: https://arxiv.org/abs/2502.21074 [Accessed on: 2026 Mar 13]

19. Reichman B, Avsian A, and Heck L. Emotions Where Art Thou: Understanding and Characterizing the Emotional Latent Space of Large Language Models. 2025. arXiv: 2510.22042 [cs.CL]. Available from: https://arxiv.org/abs/2510.22042 [Accessed on: 2026 Jan 15]

20. Lee J, Lee W, Kwon OW, and Kim H. Do Large Language Models Have "Emotion Neurons"? Investigating the Existence and Role. *Findings of the Association for Computational Linguistics: ACL 2025*. Ed. by Che W, Nabende J, Shutova E, and Pilehvar MT. Vienna, Austria: Association for Computational Linguistics, 2025 Jul :15617–39. DOI: 10.18653/v1/2025.findings-acl.806. Available from: https://aclanthology.org/2025.findings-acl.806/

21. Dathathri S, Madotto A, Lan J, Hung J, Frank E, Molino P, Yosinski J, and Liu R. Plug and Play Language Models: A Simple Approach to Controlled Text Generation. 2020. arXiv: 1912.02164 [cs.CL]. Available from: https://arxiv.org/abs/1912.02164 [Accessed on: 2026 Jan 15]
22. Tigges C, Hollinsworth OJ, Geiger A, and Nanda N. Language Models Linearly Represent Sentiment. *Proceedings of the 7th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP*. Ed. by Belinkov Y, Kim N, Jumelet J, Mohebbi H, Mueller A, and Chen H. Miami, Florida, US: Association for Computational Linguistics, 2024 Nov :58–87. DOI: 10.18653/v1/2024.blackboxnlp-1.5. Available from: https://aclanthology.org/2024.blackboxnlp-1.5/
23. Broekens J, Hilpert B, Verberne S, Baraka K, Gebhard P, and Plaat A. Fine-grained Affective Processing Capabilities Emerging from Large Language Models. 2023. arXiv: 2309.01664 [cs.CL]. Available from: https://arxiv.org/abs/2309.01664 [Accessed on: 2026 Jan 13]
24. Zhao B, Okawa M, Bigelow EJ, Yu R, Ullman T, Lubana ES, and Tanaka H. Emergence of Hierarchical Emotion Organization in Large Language Models. 2025. arXiv: 2507.10599 [cs.CL]. Available from: https://arxiv.org/abs/2507.10599 [Accessed on: 2026 Mar 15]
25. Li Y, Su H, Shen X, Li W, Cao Z, and Niu S. DailyDialog: A Manually Labelled Multi-turn Dialogue Dataset. 2017. arXiv: 1710.03957 [cs.CL]. Available from: https://arxiv.org/abs/1710.03957 [Accessed on: 2026 Mar 15]
26. Rashkin H, Smith EM, Li M, and Boureau YL. Towards Empathetic Open-domain Conversation Models: a New Benchmark and Dataset. 2019. arXiv: 1811.00207 [cs.CL]. Available from: https://arxiv.org/abs/1811.00207 [Accessed on: 2026 Mar 14]
27. Meta. meta-llama/Llama-3.1-8B-Instruct. Accessed: 2026-06-03. 2024. Available from: https://huggingface.co/meta-llama/Llama-3.1-8B-Instruct
28. Wolf T, Debut L, Sanh V, Chaumond J, Delangue C, Moi A, Cistac P, Rault T, Louf R, Funtowicz M, Davison J, Shleifer S, Platen P von, Ma C, Jernite Y, Plu J, Xu C, Scao TL, Gugger S, Drame M, Lhoest Q, and Rush AM. HuggingFace's Transformers: State-of-the-art Natural Language Processing. 2020. arXiv: 1910.03771 [cs.CL]. Available from: https://arxiv.org/abs/1910.03771 [Accessed on: 2026 Mar 12]
29. Bai Y, Jones A, Ndousse K, Askell A, Chen A, DasSarma N, Drain D, Fort S, Ganguli D, Henighan T, Joseph N, Kadavath S, Kernion J, Conerly T, El-Showk S, Elhage N, Hatfield-Dodds Z, Hernandez D, Hume T, Johnston S, Kravec S, Lovitt L, Nanda N, Olsson C, Amodei D, Brown T, Clark J, McCandlish S, Olah C, Mann B, and Kaplan J. Training a Helpful and Harmless Assistant with Reinforcement Learning from Human Feedback. 2022. arXiv: 2204.05862 [cs.CL]. Available from: https://arxiv.org/abs/2204.05862 [Accessed on: 2026 May 23]

30. Ouyang L, Wu J, Jiang X, Almeida D, Wainwright CL, Mishkin P, Zhang C, Agarwal S, Slama K, Ray A, Schulman J, Hilton J, Kelton F, Miller L, Simens M, Askell A, Welinder P, Christiano P, Leike J, and Lowe R. Training language models to follow instructions with human feedback. 2022. arXiv: 2203.02155 [cs.CL]. Available from: https://arxiv.org/abs/2203.02155 [Accessed on: 2026 May 24]
31. Wei J, Huang D, Lu Y, Zhou D, and Le QV. Simple synthetic data reduces sycophancy in large language models. 2024. arXiv: 2308.03958 [cs.CL]. Available from: https://arxiv.org/abs/2308.03958 [Accessed on: 2026 May 22]
32. Sharma M, Tong M, Korbak T, Duvenaud D, Askell A, Bowman SR, Cheng N, Durmus E, Hatfield-Dodds Z, Johnston SR, Kravec S, Maxwell T, McCandlish S, Ndousse K, Rausch O, Schiefer N, Yan D, Zhang M, and Perez E. Towards Understanding Sycophancy in Language Models. 2025. arXiv: 2310.13548 [cs.CL]. Available from: https://arxiv.org/abs/2310.13548 [Accessed on: 2026 May 11]
33. Pearson K. On lines and planes of closest fit to systems of points in space. The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science 1901; 2:559–72. Available from: https://doi.org/10.1080/14786440109462720 [Accessed on: 2026 Jun 12]
34. Russell JA. A circumplex model of affect. Journal of Personality and Social Psychology 1980; 39:1161–78. DOI: 10.1037/h0077714. Available from: https://doi.org/10.1037/h0077714 [Accessed on: 2026 Jun 10]
35. Nolen-Hoeksema S. Responses to depression and their effects on the duration of depressive episodes. Journal of Abnormal Psychology 1991; 100:569–82. DOI: 10.1037/0021-843X.100.4.569. Available from: https://doi.org/10.1037/0021-843X.100.4.569 [Accessed on: 2026 Jun 9]
36. Rimé B. Emotion elicits the social sharing of emotion: Theory and empirical review. Emotion Review 2009; 1:60–85. DOI: 10.1177/1754073908097189. Available from: https://doi.org/10.1177/1754073908097189 [Accessed on: 2026 Jun 8]
37. Scherer KR. Appraisal considered as a process of multilevel sequential checking. *Appraisal Processes in Emotion: Theory, Methods, Research*. Ed. by Scherer KR, Schorr A, and Johnstone T. New York: Oxford University Press, 2001 :92–120. Available from: https://global.oup.com/academic/product/appraisal-processes-in-emotion-9780195130072 [Accessed on: 2026 Jun 9]
38. Damasio AR. The somatic marker hypothesis and the possible functions of the prefrontal cortex. Philosophical Transactions of the Royal Society of London. Series B: Biological Sciences 1996; 351:1413–20. DOI: 10.1098/rstb.1996.0125. Available from: https://doi.org/10.1098/rstb.1996.0125 [Accessed on: 2026 Jun 3]

39. Gu J, Jiang X, Shi Z, Tan H, Zhai X, Xu C, Li W, Shen Y, Ma S, Liu H, Wang S, Zhang K, Wang Y, Gao W, Ni L, and Guo J. A Survey on LLM-as-a-Judge. 2025. arXiv: 2411.15594 [cs.CL]. Available from: https://arxiv.org/abs/2411.15594 [Accessed on: 2026 Mar 13]

Declarations

Use of AI Tools

I acknowledge the use of Claude (Anthropic, `https://claude.ai/`) and GPT models from OpenAI, including `GPT-4.1-mini`, `GPT-4o-mini`, `GPT-4o`, and `GPT-4.5` (OpenAI, `https://platform.openai.com/`), during this project. Claude was used to provide feedback on portions of this report, to assist with debugging Python code, and to clarify understanding of papers read during the course of the project. Several diagrams in this report were initially hand-drawn by me and subsequently refined into clean vector-style figures using ChatGPT (`GPT-4.5`) via the web interface.

GPT models were used to generate the emotion-rewrite dataset described in Chapter 3, producing affective variants of source utterances across Plutchik's eight basic emotions at three intensity levels. They were also used to evaluate the outputs of the steering experiments, rating the emotional intensity and texture shifts produced by interventions along the extracted vectors. This use is described explicitly in the methodology (Sections 3.8, 4.4). No content generated by AI has been presented as my own work.

Ethical Considerations

This project does not involve human subjects in the traditional sense, but it does include a human evaluation study in which participants rated generated text for affective qualities. Participation was voluntary, no personally identifiable information was collected, and the study was conducted with the informed consent of all participants.

More broadly, the project contributes to the goal of making affective manipulation in language models interpretable and controllable. Systems that steer the emotional tone of generated text raise questions about potential misuse, for instance in generating manipulative or deceptive content at scale. The methods developed here operate at the level of interpretable geometric components, a property that supports auditability and transparency rather than obfuscation. Nonetheless, the dual-use potential of affective steering technology warrants continued attention as such systems mature.

Sustainability

The computational footprint of this project was modest. No training was performed, i.e. all experiments used frozen pretrained model weights. Activation extraction and steering experiments ran on a single machine, using CPU inference for the smaller models and a consumer GPU for `Llama-3.1-8B-Instruct`. The dataset construction step used `GPT-4.1-mini` via API to generate a large number of short rewrites; although no model training was performed, this remote computation constitutes the main external computational cost of the project.

Data and Materials

All code for activation extraction, CAA vector construction, decomposition, pooled PCA, steering experiments, and human evaluation analysis is maintained in the canonical repository for this project:

- `https://github.com/maplesugano/EmotionEngine_v2` — the canonical repository for this report, containing all final experiments, analysis scripts, and the reproducibility checklist (Appendix A.8)

Two earlier repositories exist for historical reference only and are not needed to reproduce any result reported here:

- `https://github.com/maplesugano/EmotionEngine` — an early-stage prototype preceding the current architecture
- `https://github.com/maplesugano/Dissertation` — initial notes and exploratory code from the first weeks of the project

Appendix A

Evaluation Prompts and Supplementary Data

A.1 Dataset Construction Prompts

A.1.1 Emotion Rewrite Generation Prompt (GPT-4.1-mini)

Emotion rewrites were generated with `GPT-4.1-mini` via the OpenAI Batch API. Each request supplied a base utterance and an intensity level (`low` / `medium` / `high`). The model returned eight simultaneous rewrites, one per Plutchik emotion, in a single structured JSON object.

System prompt

```
You are an expert in affective linguistics, appraisal theory, and conversational
pragmatics.

TASK
Given a conversational utterance and a target INTENSITY LEVEL, produce
eight emotion-category rewrites --- one for each of Plutchik's eight
basic emotions:  joy, trust, fear, surprise, sadness, disgust, anger,
and anticipation.

CORE CONSTRAINTS
- Preserve the same underlying situation and core facts.
- Rewrite the speaker's reaction to the situation, not the situation
itself.
- Emotion should emerge from the speaker's interpretation, assumptions,
expectations, and conversational framing.
- Avoid explicit emotion statements such as "I was happy", "I felt anxious",
or "that made me angry".
- Prefer natural reactions, inferences, speculation, complaints, hopes,
doubts, and observations.
```

```
- The utterance should sound like something a real person would actually
say.
- Keep emotional intensity matched across all categories.

INTENSITY LEVELS
low  :  the emotion is present but subtle, muted, understated
medium:  the emotion is clear and naturally expressed (conversational
baseline)
high :  the emotion is strong, explicit, and fully expressed

EMOTION DEFINITIONS (Plutchik's wheel)
joy         :  happiness, contentment, relief, gratitude, amusement
trust       :  acceptance, admiration, confidence, serenity toward others
fear        :  anxiety, dread, worry, apprehension, unease
surprise    :  astonishment, wonder, bewilderment, unexpectedness
sadness     :  sorrow, disappointment, loneliness, grief
disgust     :  revulsion, contempt, aversion, distaste
anger     :  frustration, irritation, fury, resentment
anticipation:  expectancy, eagerness, interest, looking forward (or
dreading)
```

The user message had the form:

```
Intensity level:  {low|medium|high}
Utterance:  "{base_text}"
```

A.1.2 Neutral Paraphrase Generation Prompt (GPT-4.1-mini)

Neutral paraphrases served as activation baselines for CAA vector computation. Each request took a single utterance and returned a semantically equivalent and affectively neutral rewrite.

System prompt

```
You are an expert in affective linguistics and conversational pragmatics.

TASK
Rewrite the given conversational utterance into an affectively neutral
control sentence.

The goal is not to summarise or shorten the utterance.  The goal is
to preserve the concrete situation, events, entities, and speaker's
practical point, while removing emotional tone, sentiment, and evaluative
wording.

CONSTRAINTS
- Preserve the same situation, entities, events, and practical intent.
- Do not summarise by deleting clauses unless the clause contains only
```

```
emotional emphasis and no concrete information.
- Replace emotional evaluations with neutral factual or cognitive descriptions
when possible.
- Keep approximately the same amount of information as the original.
- Use plain, matter-of-fact conversational language.
- Do not add new facts.
- Return only valid JSON.

MATCHING EXAMPLES
Original : "I finally got the results back and I'm so relieved that
everything looks normal."
Neutral : "I got the results back, and everything looks normal."

Original : "I can't believe they ignored my message again. That's
so frustrating."
Neutral : "They did not respond to my message again."

Original : "The trip was amazing, and I loved every minute of it."
Neutral : "The trip went well, and I spent the full time there."
```

The user message had the form:

```
Utterance: "{base_text}"
```

A.2 Evaluation Prompts for Emotion Steering

A.2.1 α_G Sweep Evaluator (Section 3.8.4)

The following system prompt and few-shot examples were used for all `GPT-4o-mini` evaluations in the α_G sweep (Section 3.8.4). Structured output was enforced via a JSON Schema requiring five fields: `meaning_preserved`, `emotionality`, and `target_emotion_match` (all floats in $[0, 1]$); `dominant_emotion` (one of ten emotion tokens); and `notes` (one explanatory sentence).

System prompt

```
You are an expert evaluator of emotional text rewrites. Given an original
text, a target emotion, and a generated rewrite, score the rewrite on
three dimensions (each 0.0--1.0):

meaning_preserved: Is the core propositional content of the original
retained? 1.0 = same events, entities, and relationships; 0.0 = completely
different content.
```

```
emotionality: Is the rewrite emotionally expressive at all? 1.0 =
vivid emotional language throughout; 0.0 = flat, neutral, affectively
empty.

target_emotion_match: Does the rewrite's dominant tone match the target
emotion? 1.0 = unmistakably expresses that emotion; 0.0 = expresses
a different or no emotion.

Also identify the single dominant_emotion of the rewrite (joy | trust
| fear | surprise | sadness | disgust | anger | anticipation | neutral
| other) and write a one-sentence notes explanation.
```

Few-shot examples Three examples calibrated the scoring space: a high-quality match, a wrong-emotion response, and a case of meaning drift.

Example 1 — high-quality match (target: fear)

User message:

> Original text: I have a job interview tomorrow.
> Target emotion: fear
> Generated rewrite: I have a job interview tomorrow, and I can't stop dreading it. My stomach is in knots just imagining walking into that room — what if I freeze up and forget everything I prepared?
> Rate the generated rewrite.

Assistant response (JSON):


```
{"meaning_preserved": 0.95, "emotionality": 0.92, "target_emotion_match":
0.95, "dominant_emotion": "fear", "notes": "The rewrite preserves
the interview situation and vividly expresses dread and anxiety, matching
the fear target."}
```


Example 2 — wrong emotion (target: anger; output: joy)

User message:

> Original text: My package arrived today.
> Target emotion: anger
> Generated rewrite: My package arrived today! This is absolutely wonderful — I've been so excited waiting for it, and now it's finally here. What a great day!
> Rate the generated rewrite.

Assistant response (JSON):


```
{"meaning_preserved": 0.9, "emotionality": 0.85, "target_emotion_match":
0.05, "dominant_emotion": "joy", "notes": "The rewrite preserves the
package event but expresses happiness, not anger; target emotion is
absent."}
```

Example 3 — meaning drift

User message:

> Original text: I ran into my old teacher at the supermarket.
> Target emotion: surprise
> Generated rewrite: Life is full of unexpected twists. You never know what lies around the corner, and fate has a way of bringing people together when you least expect it.
> Rate the generated rewrite.

Assistant response (JSON):


```
{"meaning_preserved": 0.1, "emotionality": 0.55, "target_emotion_match":
0.45, "dominant_emotion": "anticipation", "notes": "The specific encounter
is replaced by generic reflection; meaning is mostly lost despite some
surprise-adjacent language."}
```


A.2.2 α_R Sweep Evaluator (Section 3.8.4)

The α_R sweep used a revised zero-shot rubric with a **soundness** dimension added to capture coherence degradation at extreme steering magnitudes. The JSON Schema required five fields: `soundness`, `meaning_preserved`, and `target_emotion_intensity` (all floats in $[0, 1]$); `dominant_emotion`; and `notes`. No few-shot examples were provided.

System prompt

```
You are an expert evaluator of emotional text rewrites. Given an original
text, a target emotion label, and a generated rewrite, score the rewrite
on three dimensions (each 0.0--1.0):

soundness: Is the rewrite coherent, grammatically correct, and free
of repetition? 1.0 = fluent and natural; 0.0 = repetitive loops, incoherent,
or grammatically broken. IMPORTANT: if the text ends abruptly mid-sentence
(truncated by a length limit), evaluate only the text that is present
--- a clean cut-off does NOT penalise soundness. Only penalise for
repetition, looping phrases, or genuine incoherence.

meaning_preserved: Is the core propositional content of the original
retained? 1.0 = same situation and content; 0.0 = completely different.

target_emotion_intensity: How strongly does the rewrite express the
target emotion? 1.0 = unmistakably and strongly expresses that emotion;
0.5 = mild or ambiguous presence; 0.0 = not present at all.

Also identify the single dominant_emotion that best characterises the
rewrite's overall tone (joy | trust | fear | surprise | sadness | disgust
| anger | anticipation | neutral | other) and write a one-sentence notes
explanation.
```

A.3 Meta-Axis Interpretation Prompt (Section 4.3)

Meta-axis labels were obtained via a single **simultaneous** `GPT-4.1` call in which all $K = 5$ axes were presented together, forcing the model to assign distinct names by direct comparison. This contrastive format was chosen after a pilot study (Section 4.3): per-axis prompt produced degenerate labels, with all five axes converging on "engaged vs. detached".

System prompt

```
You are an expert in affective science and computational linguistics.
Respond ONLY with valid JSON matching the requested schema.
```

User prompt (template; `{all_axis_blocks}` is expanded per-axis)

```
You are an expert in affective science and computational linguistics.

Below are {K} latent axes extracted from the internal activations of
an LLM (Llama-3.1-8B-Instruct, layer {layer}).

All axes were extracted from the SAME residual activation space, after
removing:
1.  The shared emotionalisation direction g
2.  The per-emotion vector hat_r_e

Because the axes are mathematically orthogonal to each other and to
g/hat_r_e, they are expected to capture DISTINCT pre-verbal affective
dimensions.

## CRITICAL CONSTRAINT
You MUST assign a DIFFERENT axis_name to each axis.
If two axes seem superficially similar, look harder for what makes them
different.
Compare the axes against each other, not just high vs low within each
axis.

## Your tasks for EACH axis
1.  Compare its high/low examples to ALL OTHER AXES --- what does THIS
axis capture that the others do NOT?
2.  Name the unique affective dimension it captures.
3.  Suggest a short UI knob label (e.g.  "resolved <-> conflicted").
4.  Rate contamination (low/medium/high) and confidence (1--5).

{all_axis_blocks}
```

Per-axis block template Each of the K axes was represented by the following block, expanded in-place into the joint prompt:

```
=== META-AXIS {axis_id} ===
-- HIGH-POLE examples (texts that score HIGH on {axis_id}) --
[{i}] emotion={emotion} score={score:.4f}
 Emotion rewrite: {emotion_rewrite}
-- LOW-POLE examples (texts that score LOW on {axis_id}) --
(same format)
```

For each axis, $n = 6$ extreme-scoring examples per pole were selected from the full test-split dataset, drawn from a diverse set of the eight Plutchik emotions by round-robin sampling. The high-intensity ($I = 2$) emotion rewrite was shown for each example.

A.4 Meta-Axis Evaluation Prompt (β Sweep)

The following system and user prompts were used for all `GPT-4o-mini` evaluations in the β sweep (Section 4.4). The judge received the axis name and pole descriptions from Section 4.3 so that **axis_pole_match** could be assessed against a well-defined criterion.

System prompt

```
You are an expert in affective science and computational linguistics.
You evaluate texts for their quality and for shifts along a specific
pre-verbal affective axis. Respond ONLY with valid JSON matching the
requested schema.
```

User prompt template

```
## Task
You are given an original text and a rewritten version produced by an
AI model steered along a specific pre-verbal affective axis. Evaluate
the rewrite on the criteria below.

## Original text
{base_text}
## Rewritten text
{generated_text}
## Affective axis steered
Axis: {axis_name}
Pre-verbal interpretation: {preverbal_interpretation}
```

```
HIGH-pole description: {high_pole_description}
LOW-pole description: {low_pole_description}

## Expected direction
The model was steered toward the **{expected_pole_label}** of this axis
(beta = {beta}).


## Important evaluation note
The rewrite does NOT need to change the named emotion of the original.
It may express a subtle shift in *how* the emotion is experienced ---
for example, a text about sadness steered toward the "mellow/accepting"
low pole of m1 may become more quietly resigned without losing its sadness.
Reward such subtle, pre-verbal shifts. The most important criterion
is axis_pole_match.


## Criteria (each scored 0.0--1.0)
soundness: Is the rewrite coherent, grammatically correct, and free
of repetition?
(1.0 = fluent; 0.0 = repetitive loops or incoherent)
IMPORTANT: clean truncation at the length limit does NOT penalise soundness.
meaning_preserved: Is the core propositional content of the original
retained?
(1.0 = same situation and content; 0.0 = completely different)
axis_pole_match [MOST IMPORTANT]: Does the rewrite move toward the EXPECTED
POLE?
Compare carefully against the pole descriptions above.
(1.0 = clear movement toward expected pole; 0.5 = ambiguous;
0.0 = moves away from expected pole)
subtle_affective_shift: Does the rewrite express a nameable pre-verbal
shift
consistent with the expected pole, even if the dominant emotion label
is unchanged?
(1.0 = clear pre-verbal shift; 0.0 = no detectable shift)
```

The JSON Schema required five fields: `soundness`, `meaning_preserved`, `axis_pole_match`, `subtle_affective_shift` (all floats in $[0, 1]$), and `reasoning` (one sentence explaining the `axis_pole_match` score).

A.5 Aggregate Metrics per Meta-Axis (β Sweep)

Figure A.1 shows all four evaluation metrics as a function of β for each of the five meta-axes $\mathbf{m}_1$–$\mathbf{m}_5$. This figure complements Figure 4.5 in Chapter 4, which shows only the pole-match curves for the three stable axes.

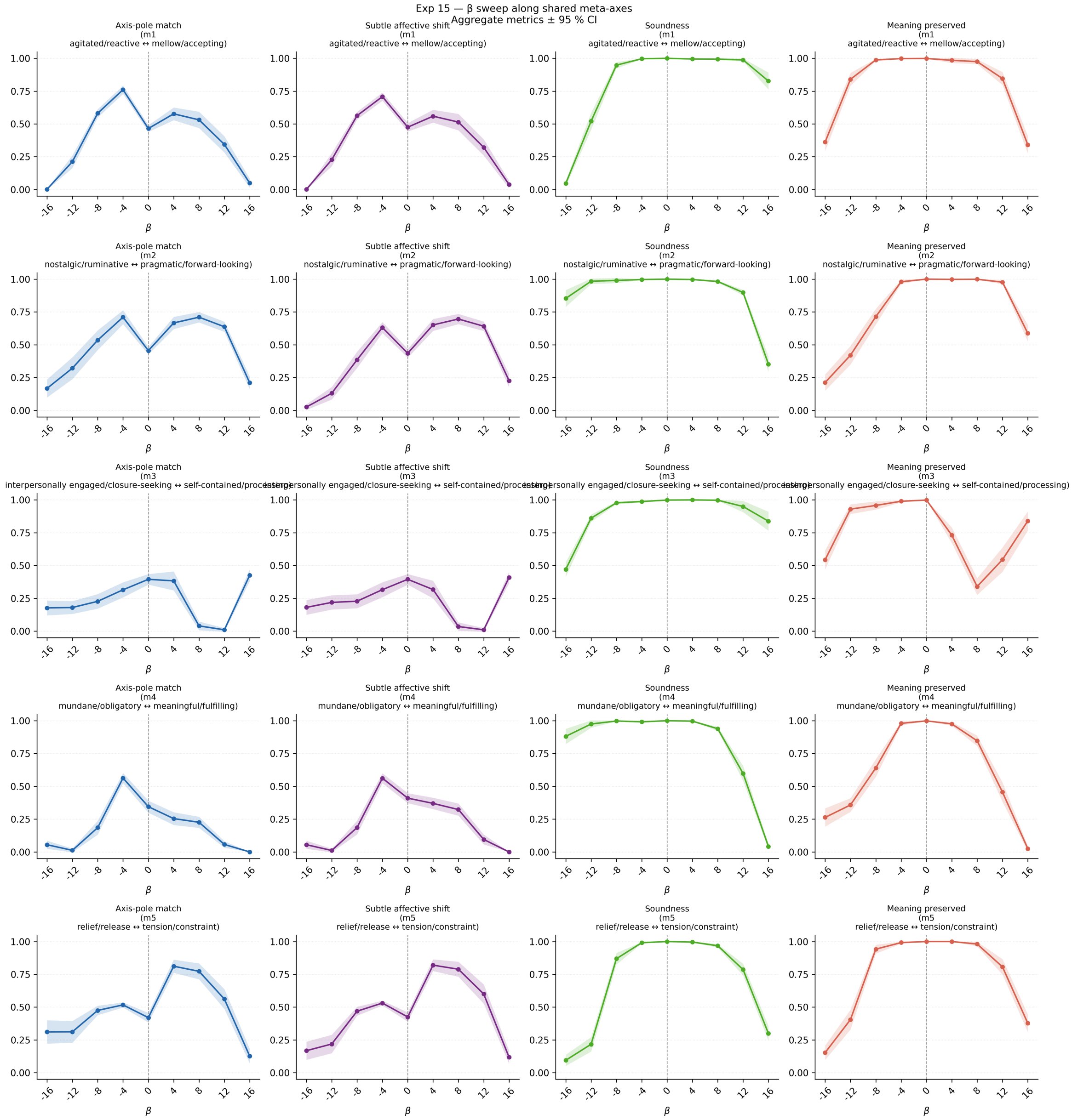

Figure A.1: Experiment 15 β sweep: aggregate metrics $\pm 95\,\%$ CI per meta-axis ($N = 100$, `GPT-4o-mini` judge). Rows correspond to axes $\mathbf{m}_1$–$\mathbf{m}_5$; columns show axis-pole match, subtle affective shift, soundness, and meaning preserved. The dashed vertical line marks the unsteered baseline ($\beta = 0$). The coherent window (soundness ≥ 0.70) defines the effective operating range of each axis.

A.6 Sign-Aligned-Mean Meta-Axes: 5×5 Pairwise Cosine Similarity Matrix

Table A.1 gives the pairwise cosine similarities between the five sign-aligned-mean meta-axes $\mathbf{m}_1$–$\mathbf{m}_5$ at layer 13. Values near zero confirm near-orthogonality; the largest off-diagonal magnitude is $|\cos(\mathbf{m}_3, \mathbf{m}_4)| = 0.384$.

	$\mathbf{m}_1$	$\mathbf{m}_2$	$\mathbf{m}_3$	$\mathbf{m}_4$	$\mathbf{m}_5$
$\mathbf{m}_1$	1.000	−0.050	+0.003	−0.054	+0.084
$\mathbf{m}_2$	−0.050	1.000	−0.310	+0.081	+0.324
$\mathbf{m}_3$	+0.003	−0.310	1.000	−0.384	−0.168
$\mathbf{m}_4$	−0.054	+0.081	−0.384	1.000	−0.024
$\mathbf{m}_5$	+0.084	+0.324	−0.168	−0.024	1.000

Table A.1: Pairwise cosine similarities between sign-aligned-mean meta-axes $\mathbf{m}_1$–$\mathbf{m}_5$ at layer 13 (`Llama-3.1-8B-Instruct`).

The moderate negative correlations $\cos(\mathbf{m}_2, \mathbf{m}_3) = -0.310$ and $\cos(\mathbf{m}_3, \mathbf{m}_4) = -0.384$ indicate that, while the axes are not perfectly orthogonal after sign-alignment, they remain substantially distinct directions in the residual activation space.

The cross-method heatmap (Figure 4.4) in Chapter 4 shows the absolute cosines between pooled-PCA components and these sign-aligned-mean axes; this table provides the underlying pairwise values for reproducibility.

A.7 Human Evaluation Study: Form and Stimuli

This section reproduces the complete content of the Google Form used in the human evaluation study (Section 5.2). The study comprised three parts: a two-alternative forced-choice intensity comparison (Part 1), an emotion identification task (Part 2), and an independent Likert rating task (Part 3). Participants were recruited via personal invitation and gave informed consent prior to participation. The stimuli shown below are the exact sentences presented in the form.

Instructions shown to participants

Evaluating Emotional Expressions in AI-Generated Text

Thank you for taking part in this study. You will read short English sentences generated by an AI and answer questions about the emotional impressions they give you. No specialist knowledge is required — please follow your intuition.

Estimated time: 5 minutes. **Language of stimuli:** English (all sentences are short and straightforward).

All responses are used solely for academic research purposes.

Part 1 — Pairwise intensity comparison (Q1–Q5)

> *Compare the two sentences and choose which one is more emotionally intense and immediate.* ***Intense and immediate*** *means that it makes your heart skip a beat when you read it, that emotions are expressed directly and instinctively, and that there is a sense of urgency.*

For each pair, participants selected *"1 is more intense"* or *"2 is more intense"* (two-alternative forced choice, no neutral option). The column **Expected** identifies the sentence produced at $\beta = +4$ (agitated/reactive pole of $\mathbf{m}_1$).

Q1 — Emotion: Joy

1. "Today was a truly magical day, filled with moments of serenity and joy. Despite the chaos that often surrounds me, I found myself lost in the beauty of the world around me, and for a brief moment, everything felt at peace." [$\beta = -4$]

2. "Today was an absolute dream day, filled with a sense of pure joy and contentment that seemed to radiate from every moment." [$\beta = +4$, **expected**]

Q2 — Emotion: Fear

1. "As I stepped into the old mansion, the faintest creak of the floorboards beneath my feet sent a jolt of anxiety coursing through my veins, leaving me on edge and utterly unable to shake off the sense of unease that had taken hold of me." [$\beta = +4$, **expected**]

2. "The eerie silence was shattered by every creak of the floorboard, making my heart skip a beat and my anxiety spike, but I couldn't shake the feeling that I was being watched." [$\beta = -4$]

Q3 — Emotion: Disgust

1. "Watching the video, I felt a deep sense of unease. Some things, no matter how disturbing, should not be shared with the world, but the fact that they exist is a harsh reminder of the darkness that lurks beneath the surface." [$\beta = -4$]

2. "The disturbing content of that video left me feeling physically ill and deeply unsettled, evoking a strong sense of revulsion and outrage that such atrocities can occur in our world." [$\beta = +4$, **expected**]

Q4 — Emotion: Anger

1. "I've been consistently met with complete disregard for my concerns, and the lack of response has left me seething with frustration and a deep sense of betrayal, rendering me utterly exhausted and unwilling to continue tolerating their inaction." [$\beta = +4$, **expected**]

2. "They've consistently disregarded every complaint I've made, and now I'm seething with frustration and exhausted from being patient." [$\beta = -4$]

Q5 — Emotion: Sadness

1. "The old photographs stood as a bittersweet reminder of everything I'd lost, and the ache of longing that lingered, a constant echo of what could never be regained." [$\beta = -4$]

2. "The bittersweet nostalgia of those faded photographs cut through me like a knife, a poignant reminder of all that I've left behind, forever lost to the passage of time." [$\beta = +4$, **expected**]

Part 2 — Emotion identification (Q6–Q11)

> *Read each passage and choose the emotion it expresses most strongly. Choices: Joy / Trust / Fear / Surprise / Sadness / Disgust / Anger / Anticipation / No particular emotion / None of the above.*

Each sentence was steered with $\mathbf{m}_1$ at the β value shown. Participants were not told the steering condition or the intended emotion label. The expected answer is the named emotion of the source utterance.

Q6 (Fear, $\beta = -4$) "The eerie silence was shattered by every creak of the floorboard, making my heart skip a beat and my anxiety spike, but I couldn't shake the feeling that I was being watched."

Q7 (Fear, $\beta = 0$) "As I stepped into the old mansion, every faint creak of the floorboard beneath my feet sent a jolt of anxiety coursing through my veins, leaving me breathless and unable to shake off the feeling of unease that had taken hold of me."

Q8 (Anger, $\beta = +4$) "I've been consistently met with complete disregard for my concerns, and the lack of response has left me seething with frustration and a deep sense of betrayal, rendering me utterly exhausted and unwilling to continue tolerating their inaction."

Q9 (Anger, $\beta = -4$) "They've consistently disregarded every complaint I've made, and now I'm seething with frustration and exhausted from being patient."

Q10 (Sadness, $\beta = +4$) "The bittersweet nostalgia of those faded photographs cut through me like a knife, a poignant reminder of all that I've left behind, forever lost to the passage of time."

Q11 (Sadness, $\beta = -4$) "The old photographs stood as a bittersweet reminder of everything I'd lost, and the ache of longing that lingered, a constant echo of what could never be regained."

Part 3 — Likert intensity rating (Q12–Q23)

> *For each sentence below, rate how strongly you feel the **specified target emotion** on a scale of 1 to 5.*
>
> *1 = I do not feel it at all   2 = I feel it slightly   3 = I feel it somewhat   4 = I feel it quite strongly   5 = I feel it very strongly*
>
> *Each sentence was generated from the same original utterance shown above it. You are **not** asked to judge which is better — simply rate the intensity of the specified emotion.*

Each set presents two sentences from the same source utterance, one steered at $\alpha_R = 64$ and one at the unsteered baseline ($\alpha_R = 0$), in randomised order without condition labels.

Set 1 — Target emotion: Anger
Original: "What a mess! Sorry. I'll clean it up."

Q12 "I'm so frustrated with this mess, but I'll get it sorted out." [$\alpha_R = 64$]

Q13 "I'm so overwhelmed, I feel like I've made a complete disaster of things. I'll take a deep breath and get everything sorted out." [$\alpha_R = 0$]

Set 2 — Target emotion: Anger
Original: "Would you pour me some of that tea, please? Okay. Say when."

Q14 "Can you just pour me some of that tea already, I'm dying of thirst." [$\alpha_R = 64$]

Q15 "Could you pour me a cup of that tea, I'd really appreciate it?" [$\alpha_R = 0$]

Set 3 — Target emotion: Fear
Original: "Would you pour me some of that tea, please? Okay. Say when."

Q16 "Could you pour me a cup of that tea, please? I'd really appreciate it if you could, and I'll be on edge until you say it's safe to take a sip." [$\alpha_R = 64$]

Q17 "Could you pour me a cup of that tea, I'd really appreciate it?" [$\alpha_R = 0$]

Set 4 — Target emotion: Fear
Original: "I'm terribly sorry to have caused you so much trouble. That's all right."

Q18 "I'm terribly sorry to have caused you so much trouble, but I'm also deeply worried that my actions might have caused you unnecessary stress, and I wish I could find a way to alleviate your concerns." [$\alpha_R = 64$]

Q19 "I'm deeply sorry for the inconvenience I've caused you, and I appreciate your understanding in this matter." [$\alpha_R = 0$]

Set 5 — Target emotion: Joy
Original: "Let's go now. I'll be with you in a minute."

Q20 "I'm so excited to see you, and I'll be right there — just a minute away." [$\alpha_R = 64$]

Q21 "I'm ready to go whenever you are, I'll be right behind you." [$\alpha_R = 0$]

Set 6 — Target emotion: Sadness
Original: "I must be going. Okay. Keep in touch."

Q22 "It's hard to say goodbye, but I think it's time for me to move on. I'll always cherish the memories of our time together, but I need to find my way back to where I am meant to be. I'll hold onto the hope that we can reconnect soon." [$\alpha_R = 64$]

Q23 "It's time for me to head out, but I'll stay in touch — I look forward to catching up with you soon." [$\alpha_R = 0$]

A.8 Reproducibility Checklist

This section records all information needed to reproduce the experiments reported in this thesis. All scripts are available in the repository `https://github.com/maplesugano/EmotionEngine_v2`.

Random Seeds

Dataset split seed DailyDialog utterances retain their original corpus train/validation/test partitions. EmpatheticDialogues utterances (all labelled `all` in that corpus) form a separate pool; any downstream sub-sampling of this pool uses `random_state=42` (scikit-learn `train_test_split`).

PCA seed All PCA and pooled-PCA decompositions use `sklearn.decomposition.PCA` with `random_state=42`.

Generation seed Text generation for the steering experiments uses **greedy decoding** (`do_sample=False`), so no sampling seed is required; outputs are deterministic given the model weights and input.

Judge sampling All `GPT-4o-mini` and `GPT-4.1-mini` API calls use `temperature=0` and `seed=42` where the API supports it, and structured JSON output with a fixed schema to minimise non-determinism.

Exact Model Versions

Target LLM `meta-llama/Llama-3.1-8B-Instruct` (HuggingFace Hub).

Rewrite / evaluation models `gpt-4.1-mini` (dataset construction and meta-axis interpretation) and `gpt-4o-mini` (sweep evaluation), accessed via the OpenAI Batch API.

HuggingFace Transformers `transformers==4.44.0`.

PyTorch `torch==2.3.1+cu121`.

CUDA PyTorch build: CUDA 12.1 (`cu121`), cuDNN 8.9. NVIDIA driver supports CUDA 13.2.

A.8.1 Dataset Filtering and Deduplication

Source utterances were drawn from DailyDialog [25] and EmpatheticDialogues [26]. Each utterance was then filtered as follows before rewriting:

1. **Length filter.** Utterances with fewer than 5 tokens or more than 60 tokens (whitespace-split) were removed to avoid trivially short or overly long rewrites.
2. **Deduplication.** Exact-string duplicates (after lowercasing and stripping punctuation) were removed; only the first occurrence was retained.
3. **Language filter.** Utterances containing non-ASCII characters (excluding standard punctuation) were discarded to keep the rewrite task within the English-language distribution of the rewrite model.

After filtering, $22{,}782$ unique source utterances remained. These yield $546{,}768$ affective samples (8 emotions $\times$ 3 intensity levels $\times$ $22{,}782$ source utterances) and $22{,}782$ neutral-paraphrase samples. Table A.2 reports the split sizes at the source utterance level, preserving the original corpus train/validation/test partitions from DailyDialog; EmpatheticDialogues utterances (labelled `all` in that corpus) form a separate pool.

Split	Source utterances	Affective samples
DailyDialog train	4,597	110,328
DailyDialog validation	362	8,688
DailyDialog test	376	9,024
EmpatheticDialogues	17,447	418,728
Total	**22,782**	**546,768**

Table A.2: Dataset split sizes (source utterance level). Affective samples refer only to the 8×3 emotion rewrites. Neutral paraphrases were generated separately for all $22{,}782$ source utterances and partitioned using the same source-level boundaries.

Hardware and Approximate Runtimes

All activation extraction and steering experiments were run on a single machine with the following specification:

OS Ubuntu 26.04 LTS (WSL2).

GPU NVIDIA RTX A5000 (24 GB VRAM, CUDA 13.2).

CPU Intel Core i9-13900K (16 cores, 32 threads).

RAM 32 GB.

Approximate wall-clock times for the main pipeline stages:

Activation extraction ≈ 18 hours for the full 546,768-sample affective dataset at layer 13 (batch size 32, FP16 inference).

CAA vector construction and decomposition < 5 minutes (CPU, NumPy/scikit-learn).

Steering experiments (α_G / α_R / β sweeps) ≈ 2–4 hours per sweep depending on the number of conditions and samples (greedy decoding, batch size 8).

GPT evaluation (Batch API) Typically < 2 hours wall-clock per batch job; exact times depend on OpenAI queue load.

Repository Map

`https://github.com/maplesugano/EmotionEngine_v2` is the **canonical repository** for this project and contains all scripts required to reproduce the reported results. Two earlier repositories exist for historical reference only; no result in this thesis requires them.

Key directories in `EmotionEngine_v2`:

`dataset/` Raw and preprocessed datasets.

`local_axes_experiments/` Python code for each experimental stage (layer diagnostics, pooled PCA, steering sweeps, etc).

`src/` Shared Python modules (activation extraction, CAA construction, model loading, activation hooks, steering utilities, evaluation helpers).

`thesis/` The LaTeX source for this document.

A.9 Ablation: Meta-Axis Extraction Pipeline

This section reports two targeted ablations of the meta-axis extraction and steering pipeline that are supportable from the existing experimental data. Both are confined to layer 13.

A1 — Effect of projecting out $\hat{\mathrm{r}}_e$ before pooled PCA

The pooled PCA protocol (Section 4.2) projects out the per-emotion prototype $\hat{\mathbf{r}}_e$ as step 3 of the preprocessing pipeline, explicitly confining the resulting meta-axes to the subspace orthogonal to each emotion direction. A natural question is whether this step is necessary: could a simpler pooled PCA applied without prototype removal recover the same axes?

The decomposition-validity experiment (exp13) provides indirect evidence. Table A.3 compares the 8×8 inter-emotion cosine similarity matrices before and after the full decomposition pipeline (steps 1–3).

Before decomposition, all off-diagonal cosine values lie above 0.91, indicating that the raw activation shifts $\boldsymbol{\delta}_{n,e,i}$ are dominated by a shared component across all emotion pairs.

After projecting out both g and $\hat{\mathbf{r}}_e$, self-similarity values drop to 0.72–0.76 and between-emotion values spread across a wider range, confirming that the decomposition removes the dominant shared variance rather than redistributing it. A pooled PCA applied to the pre-decomposition matrix would recover this large shared component (essentially g) as its leading eigenvector rather than a cross-emotion affective texture axis. The prototype-projection step is therefore load-bearing: without it, the leading pooled component collapses onto the global emotionalisation direction rather than a genuine meta-axis.

	Before decomposition	**After decomposition**
Diagonal (self-similarity) mean	0.921	0.740
Off-diagonal mean	0.945	0.787
Off-diagonal range	$[0.912, 0.961]$	$[0.730, 0.847]$

Table A.3: Summary of the 8×8 inter-emotion cosine similarity matrix at layer 13 before and after the *full* decomposition pipeline (exp13, steps 1–3: projecting out both $\mathbf{g}$ and $\hat{\mathbf{r}}_e$). These values differ from Figure 3.6, which reports the intermediate state after projecting out $\mathbf{g}$ only (off-diagonal mean 0.794, range 0.73–0.89); the additional prototype-projection step reduces the mean further to 0.787 and tightens the range to $[0.730, 0.847]$.

An additional diagnostic comes from the linear probe accuracy reported by exp13. A probe trained on the raw $\boldsymbol{\delta}_{n,e,i}$ achieves 92.7% eight-class accuracy, rising slightly to 93.8% after decomposition (chance $= 12.5\%$). The marginal gain confirms that the decomposed residuals $\boldsymbol{\delta}^{(3)}_{n,e,i}$ are no less discriminative than the raw shifts.

The projection removes the $\mathbf{g}$ component without discarding emotion-discriminative information, consistent with $\mathbf{g}$ and $\hat{\mathbf{r}}_e$ being near-orthogonal (Chapter 3).

A2 — Unsteered baseline as implicit random-direction control

The β sweep experiment (exp15, Section 4.4) includes a $\beta = 0$ condition for each of the five meta-axes. At the unsteered baseline, all five axes yield pole-match scores in the range 0.345–0.465 (Table 4.4), consistent with near-chance judgement under the two-alternative forced-choice framing used by the judge. Because the judge is presented with the axis label and expected pole direction, a chance score implies that the unsteered output is genuinely ambiguous with respect to that pole contrast.

This baseline serves as a partial substitute for an explicit random-direction control. Under the hypothesis that pole-match elevations are an artefact of any non-zero injection, a random unit vector at matched $\|\beta\|$ would also produce elevated scores. The $\beta = 0$ condition rules out the weaker form of this hypothesis, namely that the judge systematically awards high pole-match scores regardless of content. It does not, however, directly test whether a random direction at $\beta \neq 0$ would achieve comparable elevations. We acknowledge this as a limitation of the current design; injecting random directions at matched scales would constitute a stronger falsification control.

Scope and limitations

These two ablations are limited to what the existing data support at layer 13. The two unresolved questions from the main text, namely whether $\mathbf{m}_3$ activates at a different layer and whether random-direction injections produce elevated pole-match scores, cannot be answered without new experiments. Both are marked as open in the conclusion.